

Lecture 7:

Random Variables and Distributions



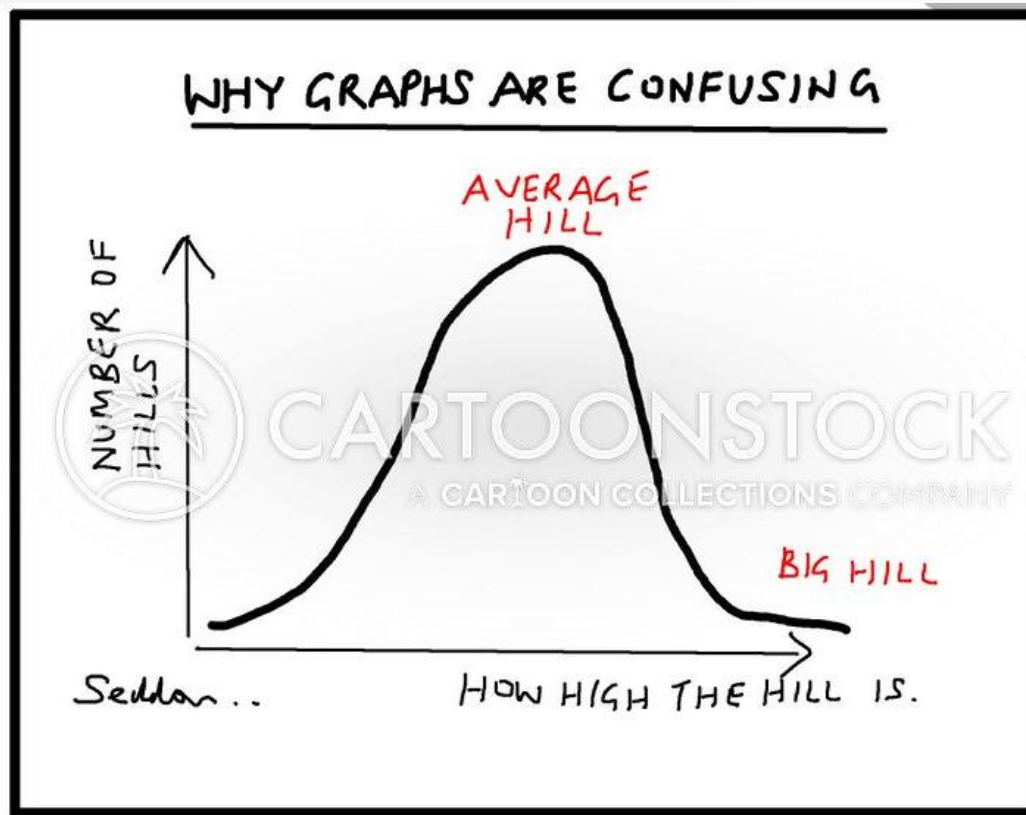
Markus Hohle

University of California, Berkeley

**Numerical Methods for
Computational Science**

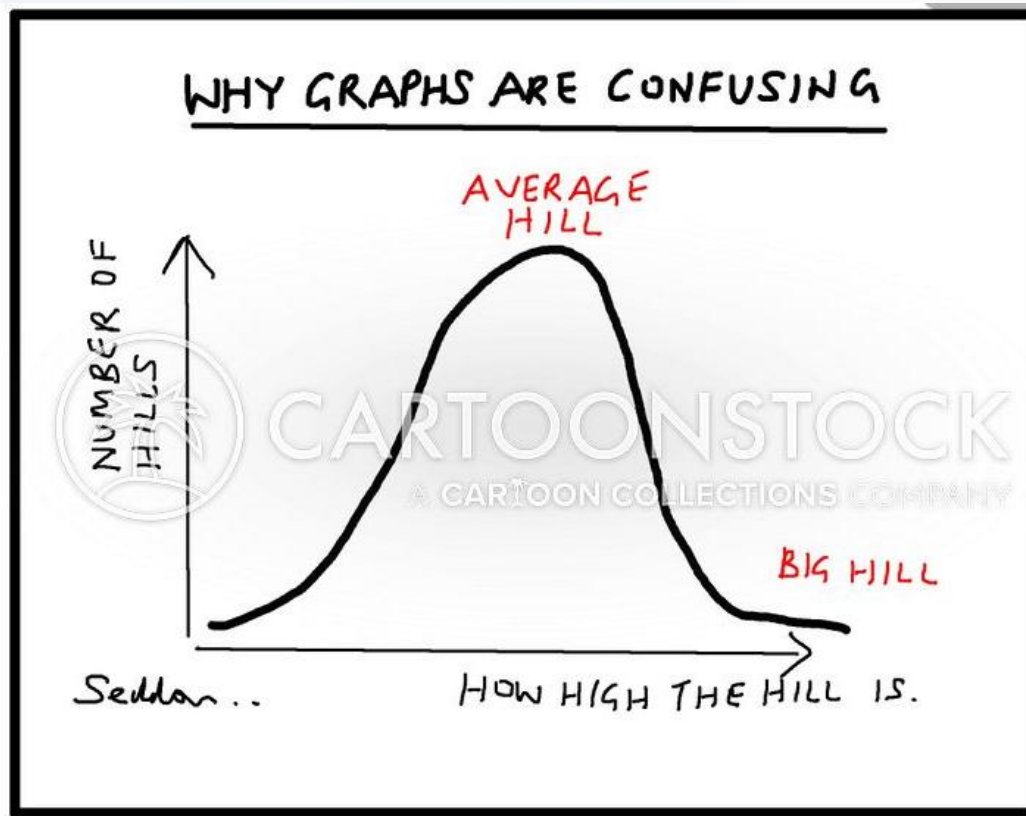
Course Map

Week 1:	Introduction to Scientific Computing and Python Libraries
Week 2:	Linear Algebra Fundamentals
Week 3:	Vector Calculus
Week 4:	Numerical Differentiation and Integration
Week 5:	Solving Nonlinear Equations
Week 6:	Probability Theory Basics
Week 7:	Random Variables and Distributions
Week 8:	Statistics for Data Science
Week 9:	Eigenvalues and Eigenvectors
Week 10:	Simulation and Monte Carlo Method
Week 11:	Data Fitting and Regression
Week 12:	Optimization Techniques
Week 13:	Machine Learning Fundamentals



Outline

- Uniform Distribution
- Binomial Distribution
- Poisson Distribution
- Normal Distribution
- Central Limit Theorem



Outline

- Uniform Distribution

- Binomial Distribution

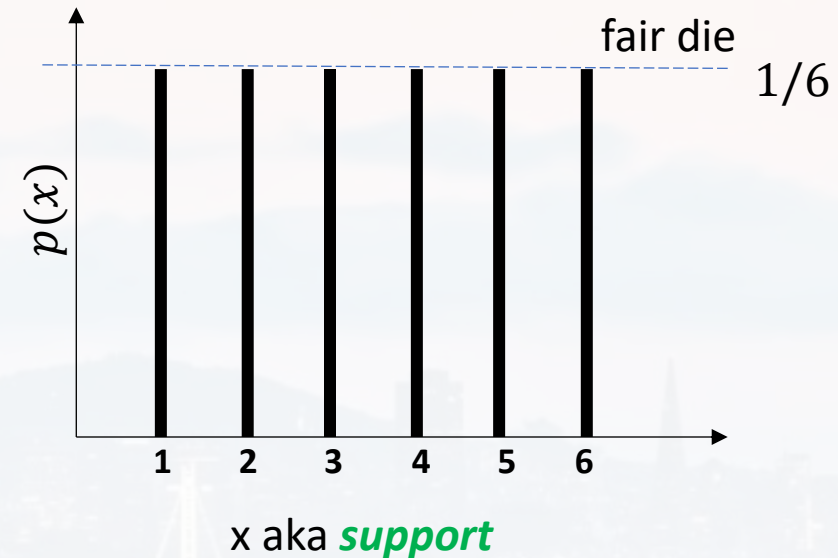
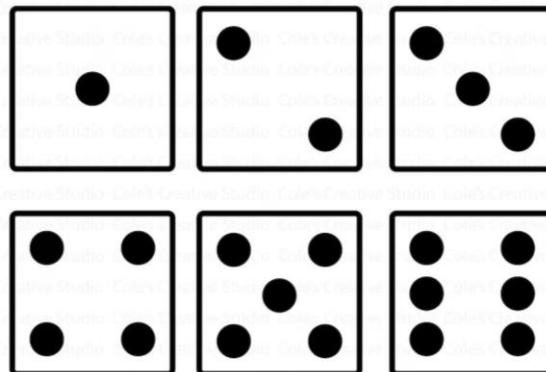
- Poisson Distribution

- Normal Distribution

- Central Limit Theorem

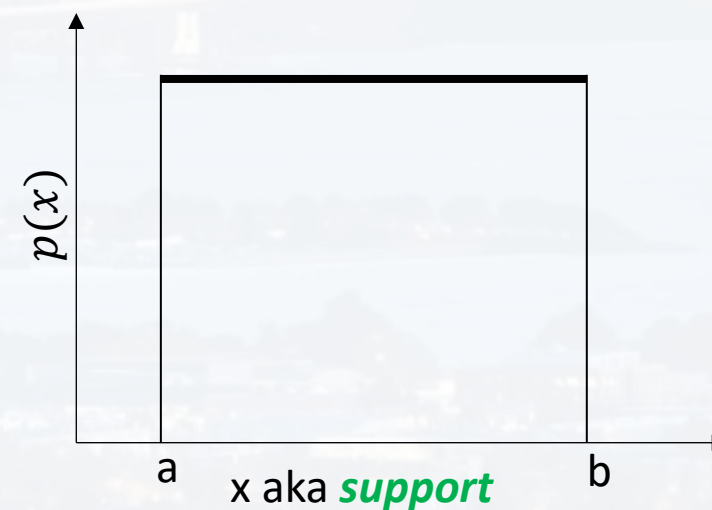
distributions

discrete (= countable)



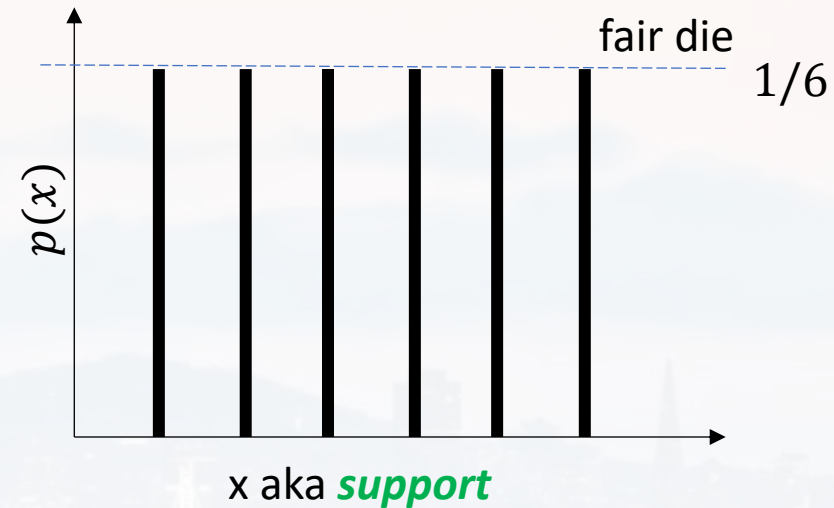
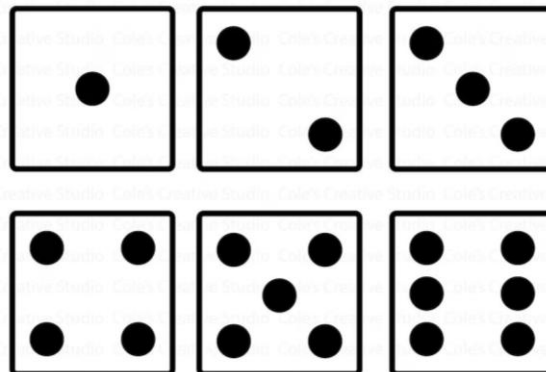
continuous

$$[a \leq x \leq b]$$



distributions

discrete (= countable)



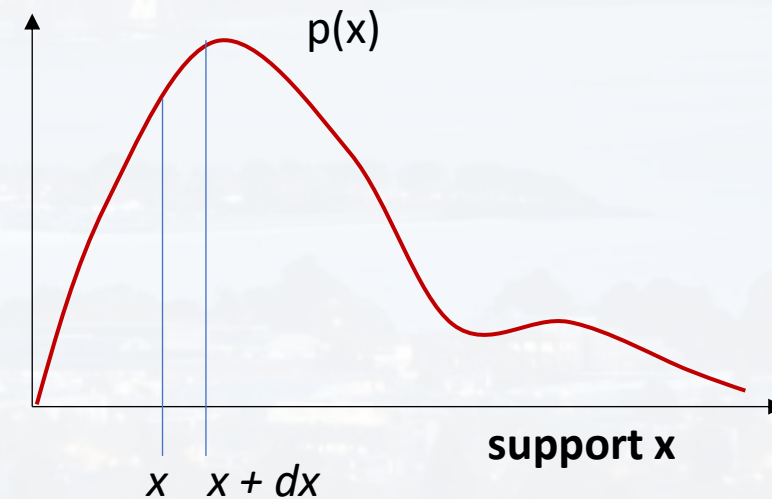
continuous

$$[a \leq x \leq b]$$

$p(x)$ doesn't make sense

$$\rightarrow p(x) dx$$

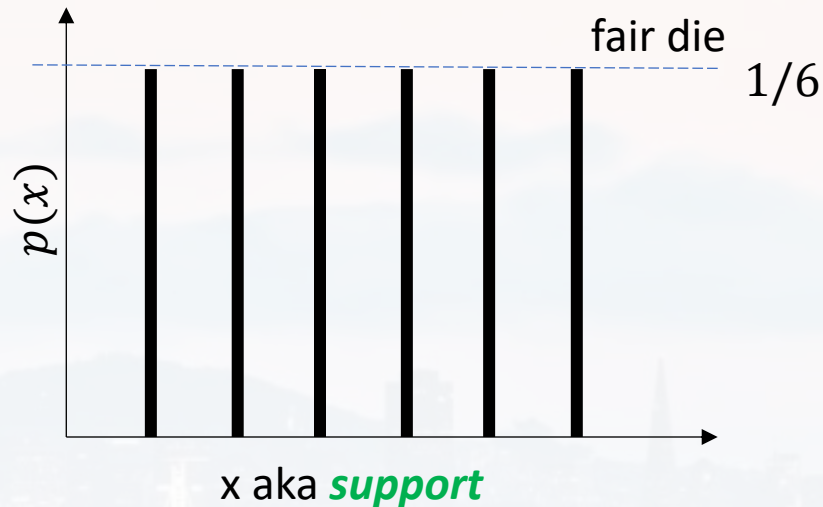
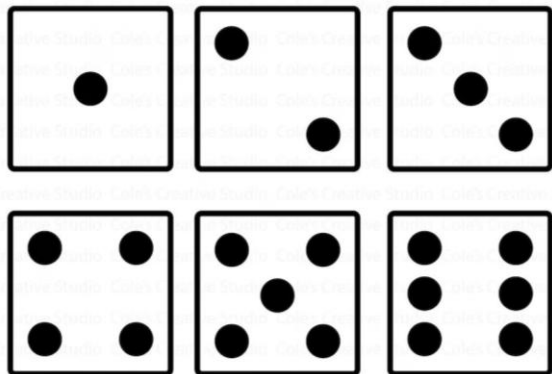
probability **d**ensity **f**unction





distributions

discrete (= countable)



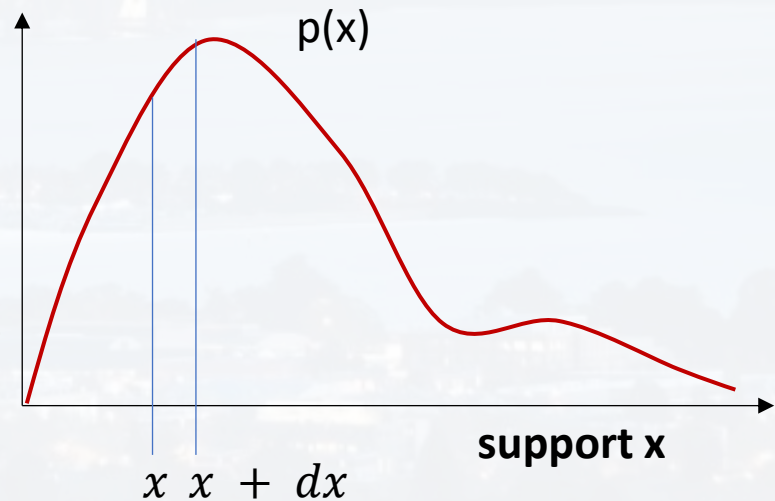
continuous

$p(x)$ doesn't make sense
 $\rightarrow p(x) dx$

probability **d**ensity **f**unction

dx defines the probability!

*What is the probability to find a person who is **EXACTLY** 6ft tall?
Depends on how **accurate** (dx) you measure!*



the mean μ

(barycenter)

the variance σ^2

(natural scatter)

discrete (= countable)

$$\mu = E(x) = \sum_i x_i p(x_i)$$

$$\sigma^2 = \text{var}(x) = \sum_i (x_i - \mu)^2 p(x_i)$$

continuous

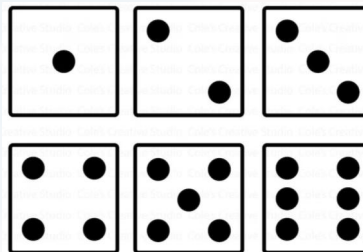
$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

uniform distribution: discrete vs continuous

the mean μ

discrete (= countable)



$$\mu = \sum_i x_i p(x_i) = \sum_{i=1}^6 i \frac{1}{6} = 3.5$$

continuous

$$[a \leq x \leq b]$$

$$\mu = \int x p(x) dx$$

$p(x) = \text{const}$
(uniform)

$$= \text{const} \int_a^b x dx = \text{const} \frac{1}{2} (b^2 - a^2)$$

2nd axiom

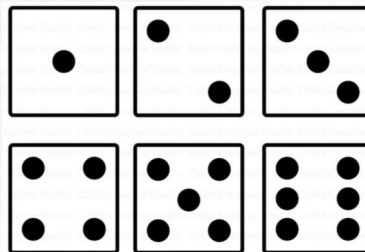
$$\int_a^b p(x) dx = 1$$

$$\text{const} \int_a^b dx = 1 \quad \rightarrow \text{const} = \frac{1}{b - a}$$

uniform distribution: discrete vs continuous

the mean μ

discrete (= countable)



$$\mu = \sum_i x_i p(x_i) = \sum_{i=1}^6 i \frac{1}{6} = \mathbf{3.5}$$

continuous

$$[a \leq x \leq b]$$

$$\mu = \int x p(x) dx = \text{const} \frac{1}{2} (b^2 - a^2)$$

$$\text{const} = \frac{1}{b - a}$$

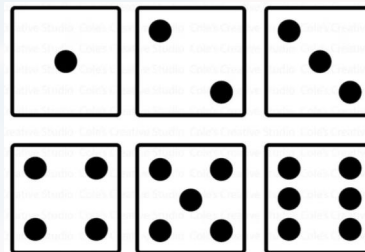
$$\boxed{\mu = \frac{1}{2} \frac{b^2 - a^2}{b - a}} = \mathbf{3.5} \text{ for } a = 1 \text{ and } b = 6$$



uniform distribution: discrete vs continuous

the variance σ^2

discrete (= countable)



$$\sigma^2 = \sum_i (x_i - \mu)^2 p(x_i) = \frac{1}{6} \sum_{i=1}^6 (i - 3.5)^2 \approx \mathbf{2.9}$$

continuous

$$[a \leq x \leq b]$$

$$\sigma^2 = \int (x - \mu)^2 p(x) dx = \frac{1}{b - a} \int_a^b (x - 3.5)^2 dx$$

$$\sigma^2 = \frac{1}{12} (b - a)^2$$

$$= 25/12 \approx \mathbf{2.1} \text{ for } a = 1 \text{ and } b = 6$$

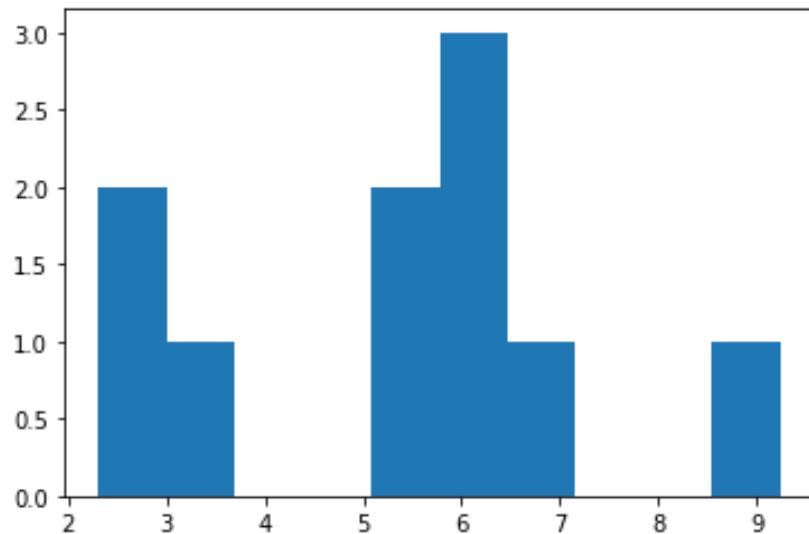
$$p(x) = \text{const}$$

plotting the pdf

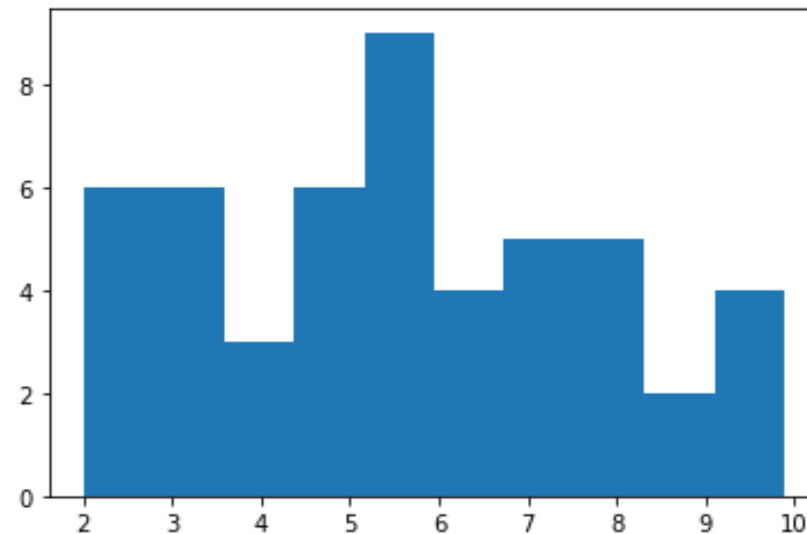
```
U = np.random.uniform(low, high, shape)  
plt.hist(U)
```

continuous support

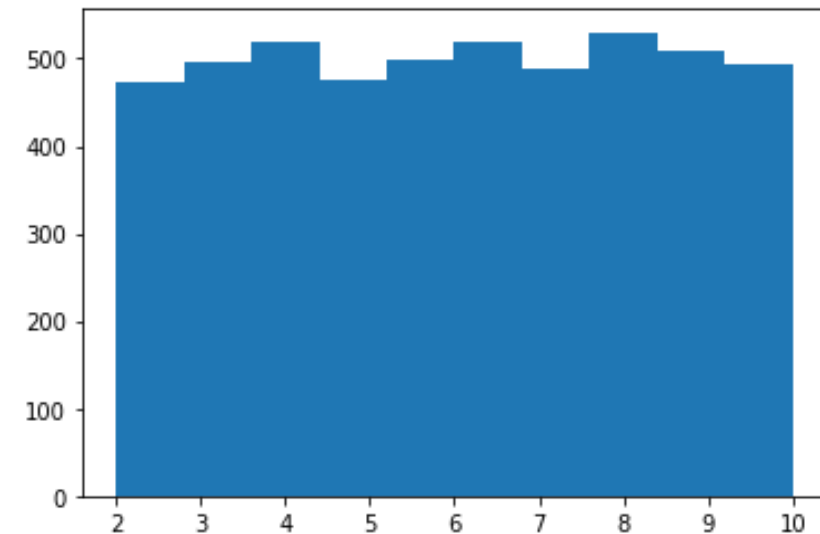
N = 10



N = 50



N = 5000



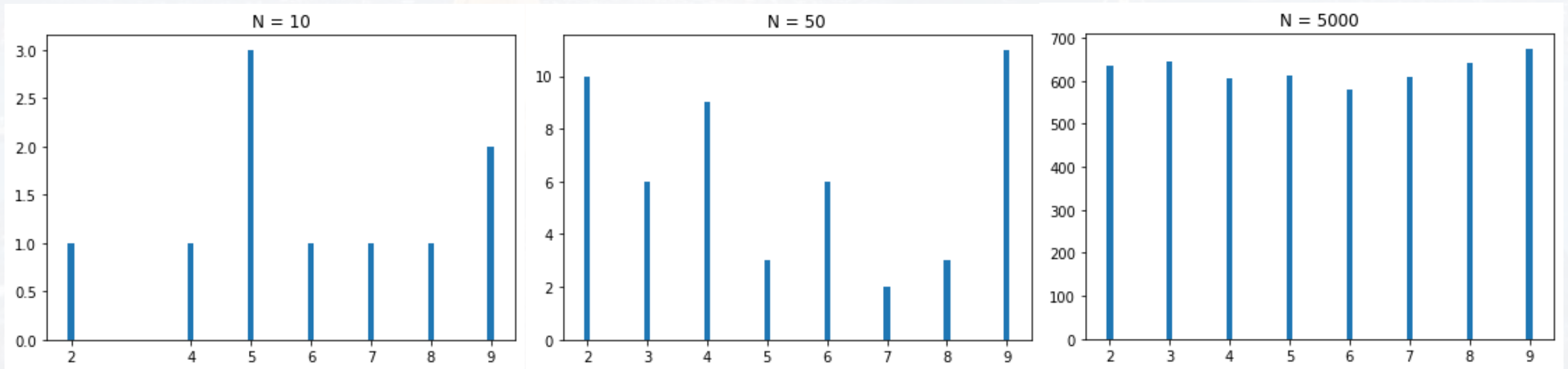
$$p(x) = \text{const}$$

plotting the pdf

```
U = np.random.randint(low, high, shape)
```

discrete support

```
labels, counts = np.unique(U, return_counts = True)
plt.bar(labels, counts, align = 'center', width = 0.1)
plt.gca().set_xticks(labels)
plt.title('N = ' + str(N))
```





Cumulative **d**ensity **f**unction

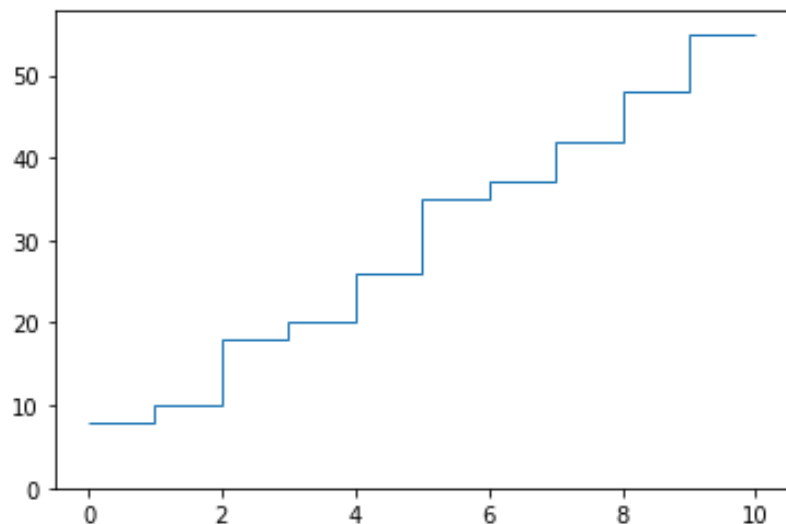
$$P(x) = \int_a^x p(x) dx$$

```
U = np.random.randint(low, high, shape)
```

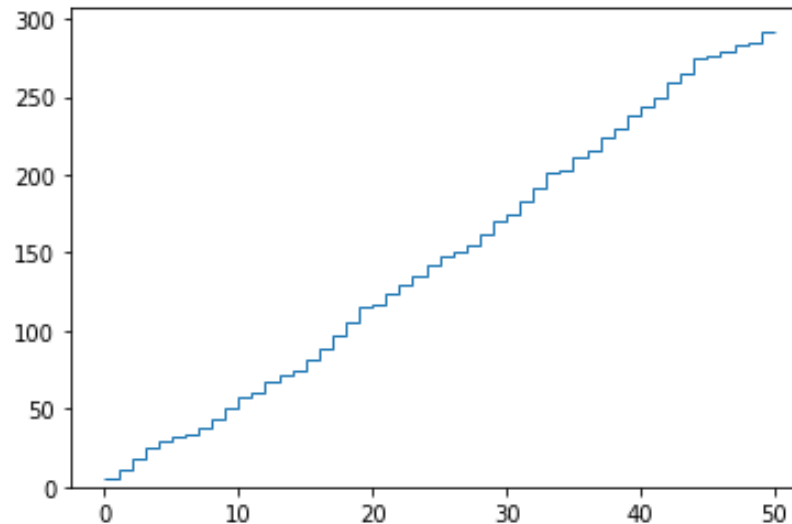
discrete support

```
C = np.cumsum(U)  
plt.stairs(C, baseline = None)  
plt.title('N = ' + str(N))
```

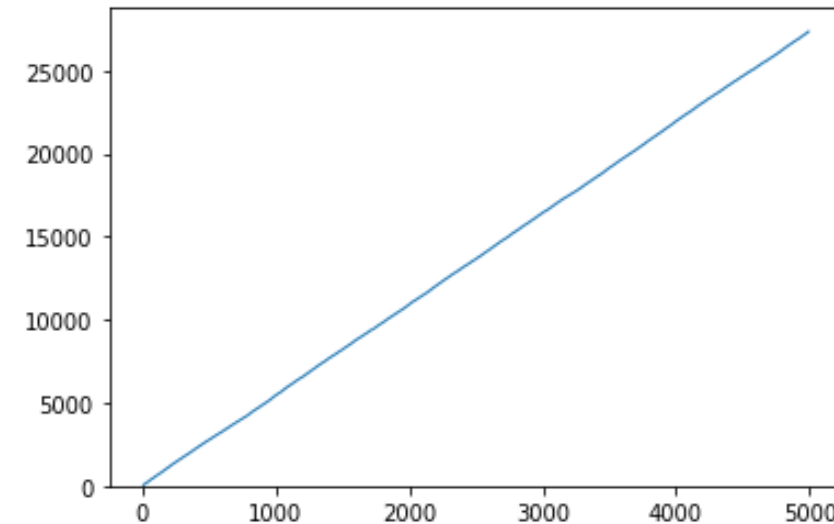
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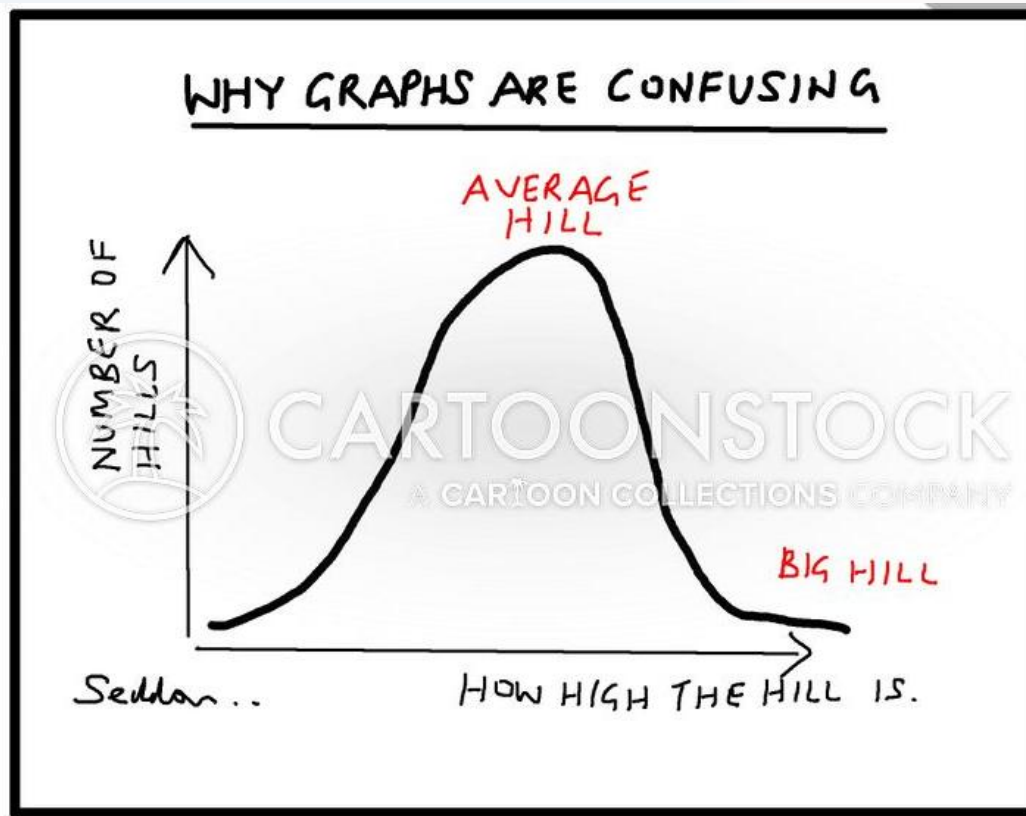


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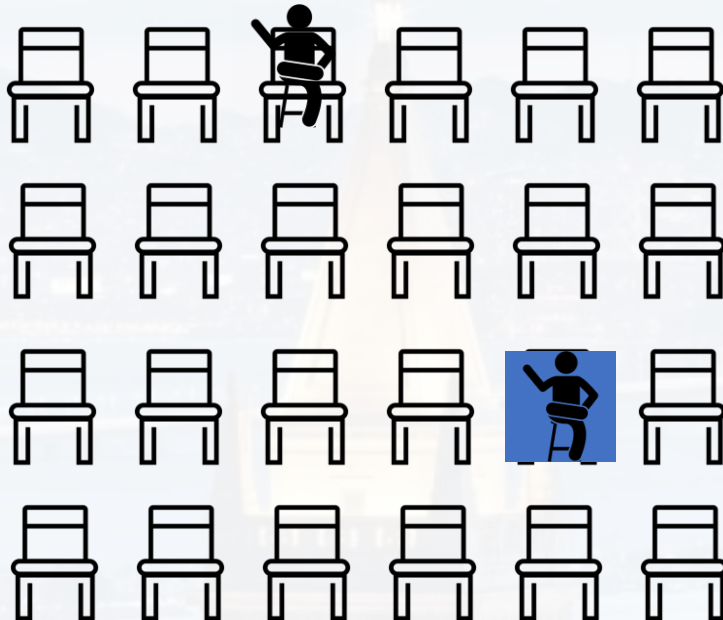
Outline

- Uniform Distribution
- **Binomial Distribution**
- Poisson Distribution
- Normal Distribution
- Central Limit Theorem

seating arrangements in a classroom

n choose k
The Binomial Distribution

- How many arrangements Ω for k students and n seats?
- How many arrangements Ω for k occupied seats among n seats?



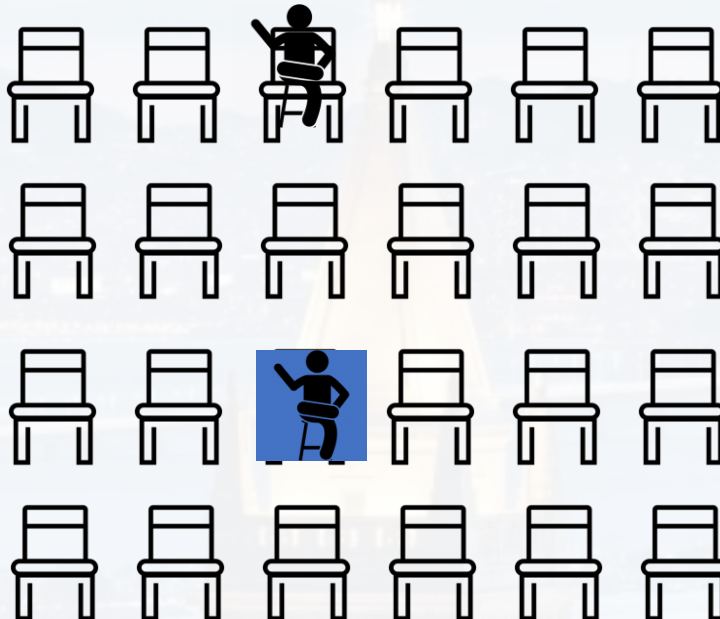
student 1: $\Omega = n$

student 2: $\Omega(2) = n - 1$

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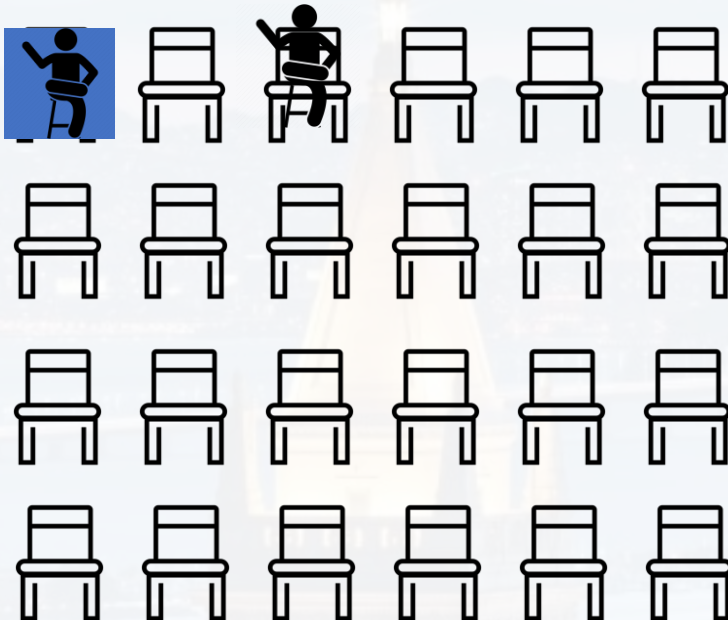
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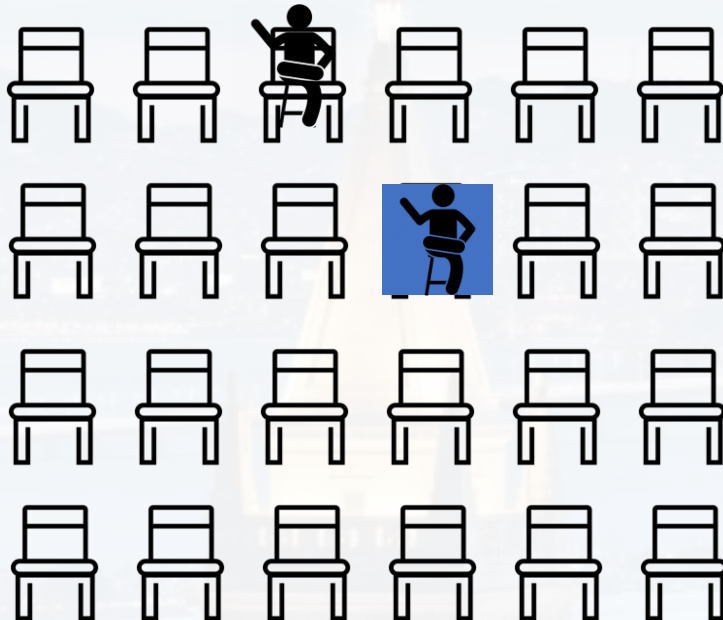
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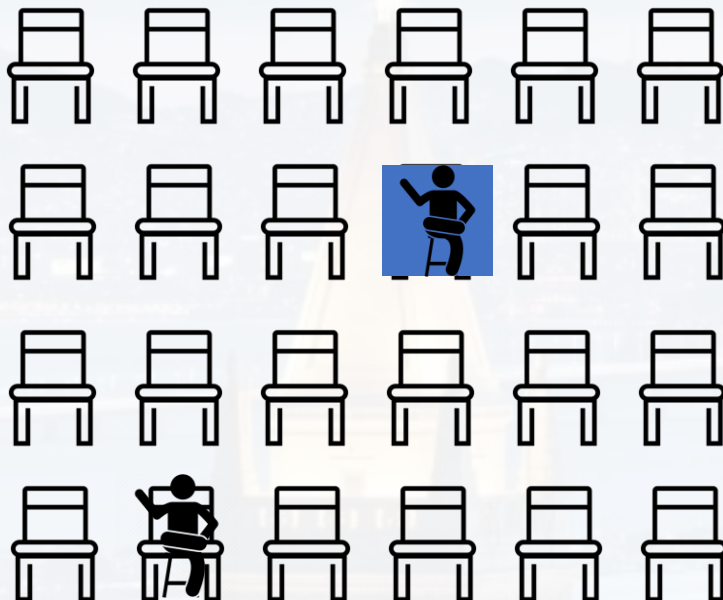
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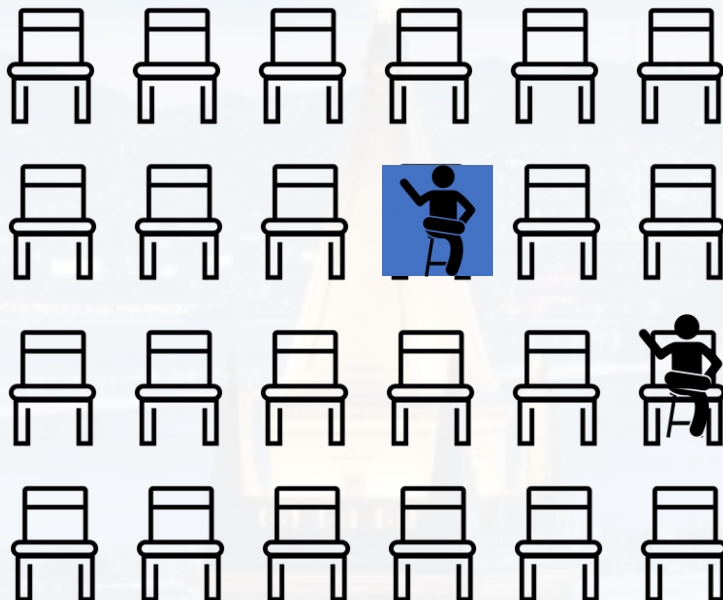
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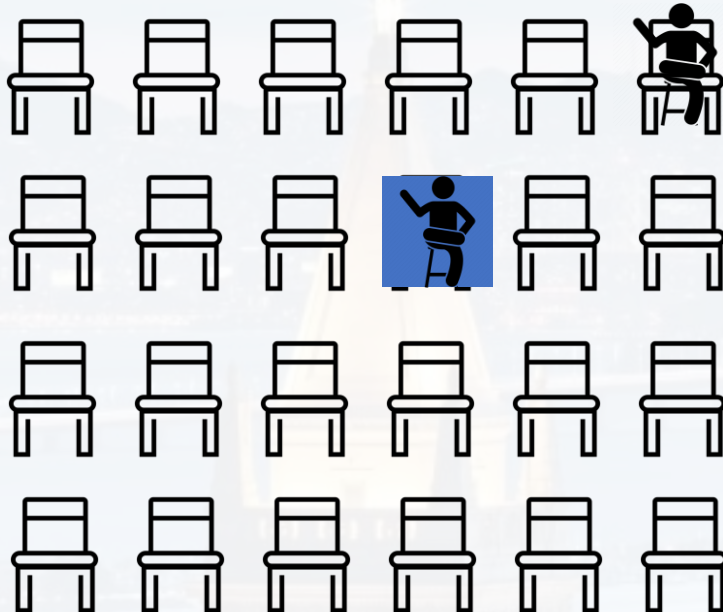
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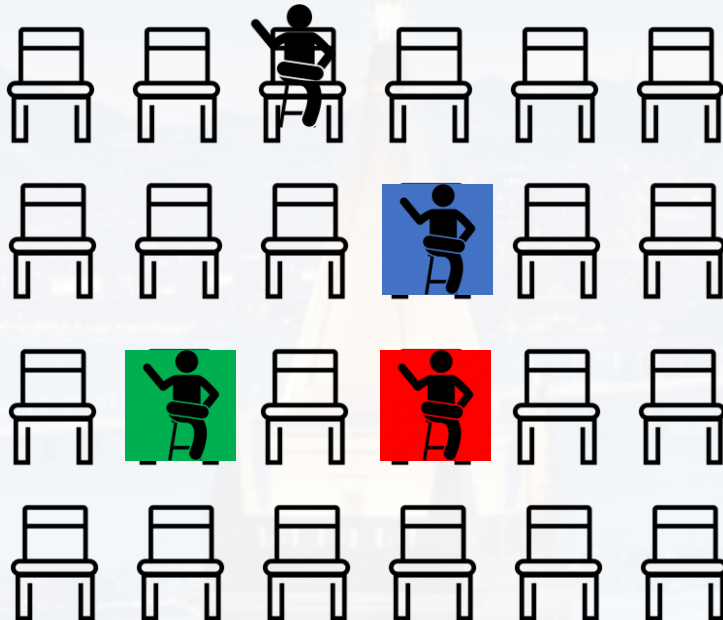
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seating arrangements in a classroom

n choose k
The Binomial Distribution

- How many arrangements Ω for k students and n seats?
- How many arrangements Ω for k occupied seats among n seats?



student 1: $\Omega = n$

student 2: $\Omega(2) = n - 1$

$$\Omega = n(n - 1)$$

student 3: $\Omega(3) = n - 2$

$$\Omega = n(n - 1)(n - 2)$$

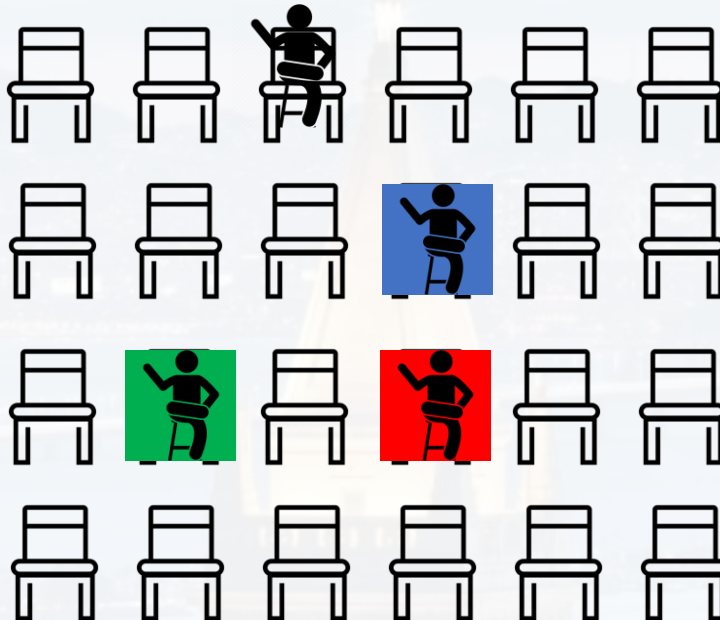
student 4: $\Omega(4) = n - 3$

$$\Omega = n(n - 1)(n - 2)(n - 3)$$

seating arrangements in a classroom

n choose k
The Binomial Distribution

- How many arrangements Ω for k students and n seats?
- How many arrangements Ω for k occupied seats among n seats?



student 1: $\Omega = n$

student 2: $\Omega = n (n - 1)$

student 3: $\Omega = n (n - 1)(n - 2)$

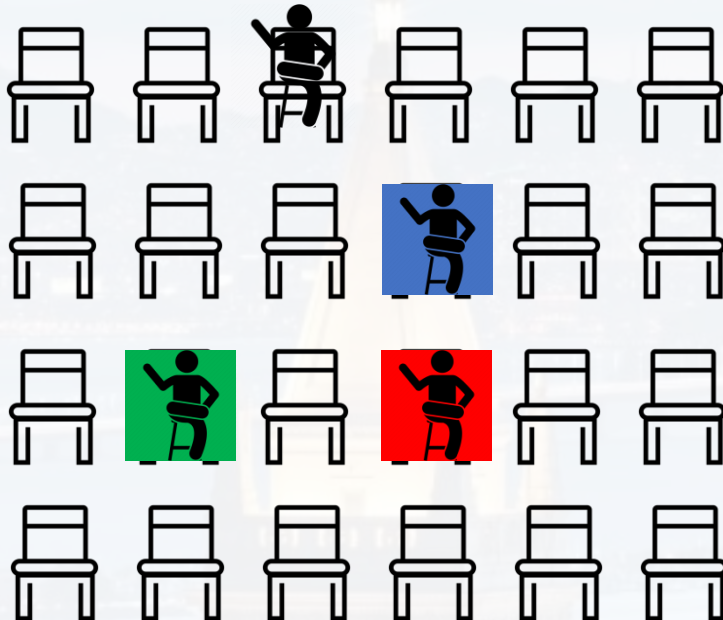
student 4: $\Omega = n (n - 1)(n - 2)(n - 3)$

student k : $\Omega = n (n - 1)(n - 2)(n - 3) \dots (n - k + 1)$

seating arrangements in a classroom

n choose k
The Binomial Distribution

- How many arrangements Ω for k students and n seats?
- How many arrangements Ω for k occupied seats among n seats?



student k :

$$\Omega = n (n - 1)(n - 2)(n - 3) \dots (n - k + 1)$$

for $k = n$:

$$\Omega = k (k - 1)(k - 2) \dots 1 = k!$$

k factorial

$$5! = 5 * 4 * 3 * 2 * 1 = 120$$

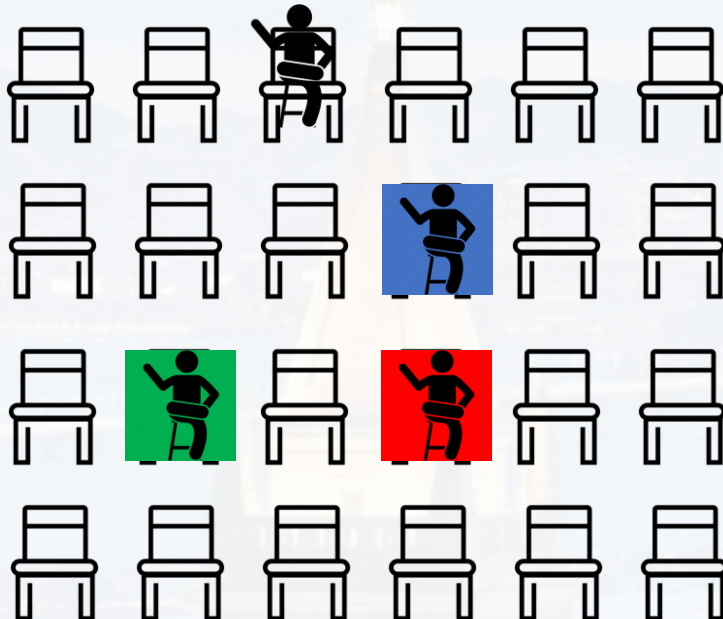
note: $0! = 1$

seating arrangements in a classroom

n choose k

The Binomial Distribution

- How many arrangements Ω for k students and n seats?
- How many arrangements Ω for k occupied seats among n seats?



student k :

$$\Omega = n(n-1)(n-2)(n-3) \dots (n-k+1)$$



interrupting at $(n-k)!$

$$\Omega = \frac{n!}{(n-k)!} = \frac{n(n-1) \dots (n-k+1)(n-k)(n-k-1) \dots 1}{(n-k)(n-k-1) \dots 1}$$

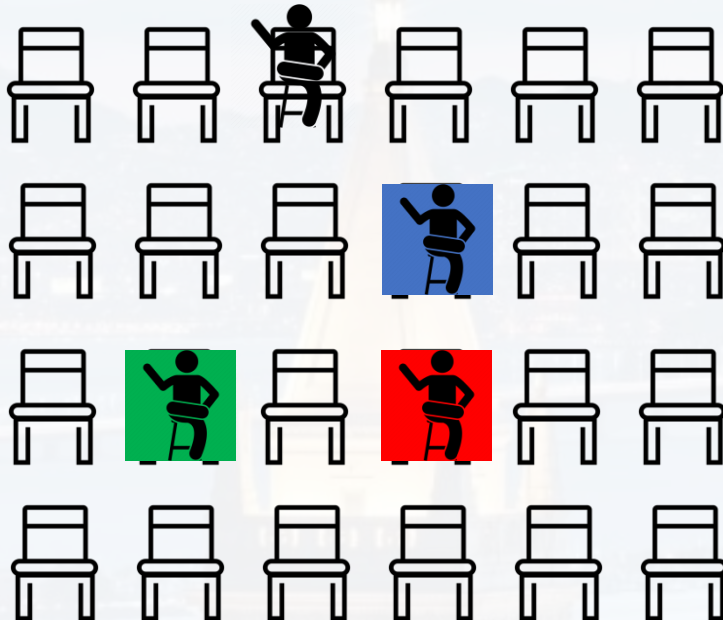
$$\Omega = \frac{n!}{(n-k)!}$$

seating arrangements in a classroom

n choose k

The Binomial Distribution

- How many arrangements Ω for k students and n seats? 😊
- How many arrangements Ω for k occupied seats among n seats?



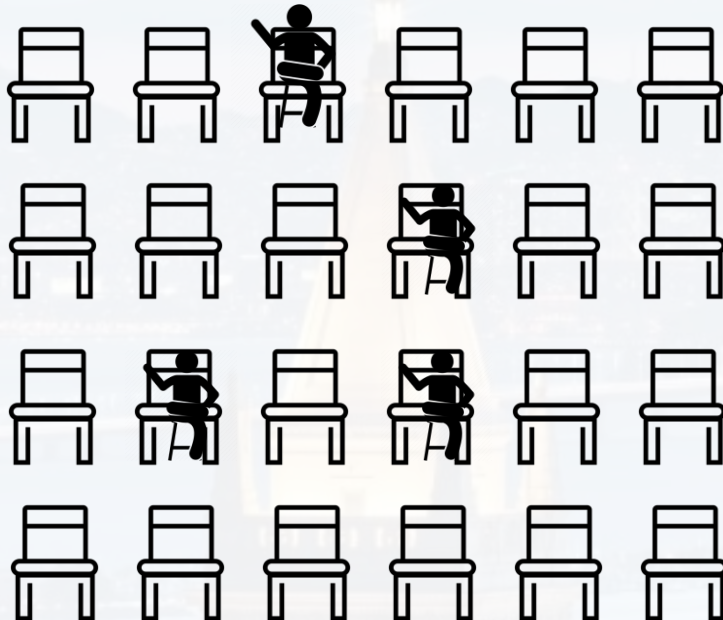
$$\Omega = \frac{n!}{(n - k)!}$$

seating arrangements in a classroom

n choose k

The Binomial Distribution

- How many arrangements Ω for k students and n seats?
- **How many arrangements Ω for k occupied seats among n seats?**



$$\Omega = \frac{n!}{(n-k)!}$$

k students on k seats

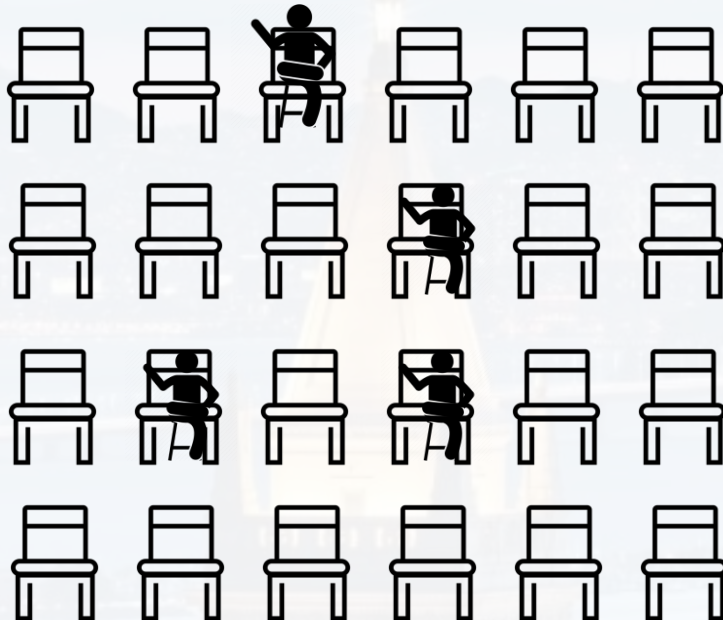
→ $k!$ arrangements, which are *indistinguishable*

$$\Omega = \frac{n!}{k! (n-k)!} = \binom{n}{k}$$

n choose k

seating arrangements in a classroom

- How many arrangements Ω for k students and n seats?
- **How many arrangements Ω for k occupied seats among n seats?**



$k!$ are *indistinguishable*

$$\Omega = \frac{n!}{k!(n-k)!} = \binom{n}{k}$$

Stirling's approximation

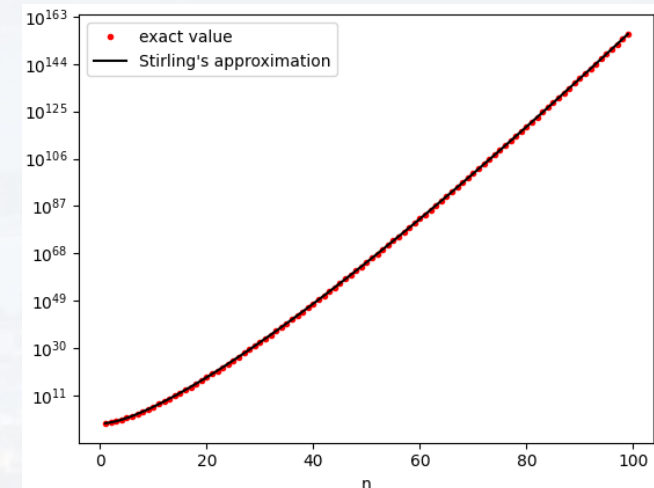
$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

n choose k

The Binomial Distribution

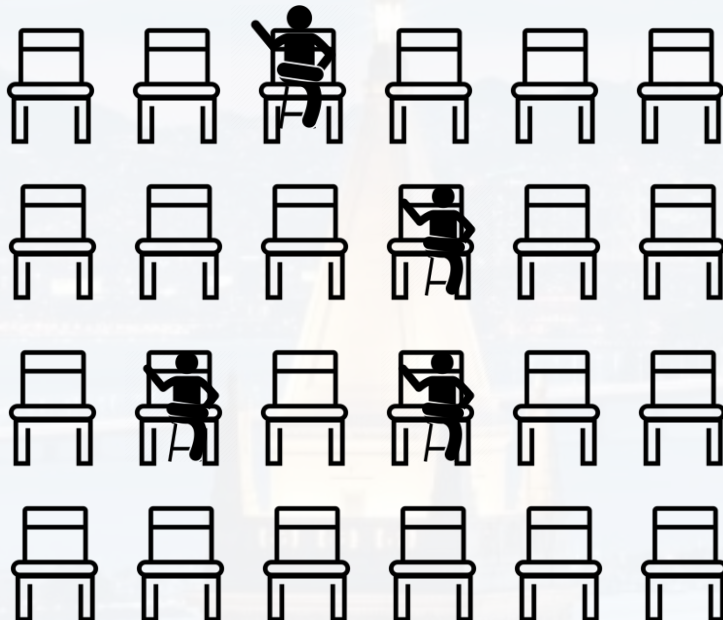
see `nchoosek.ipynb`

n choose k



seating arrangements in a classroom

- How many arrangements Ω for k students and n seats?
- **How many arrangements Ω for k occupied seats among n seats?**



$k!$ are *indistinguishable*

$$\Omega = \frac{n!}{k! (n - k)!} = \binom{n}{k}$$

Stirling's approximation

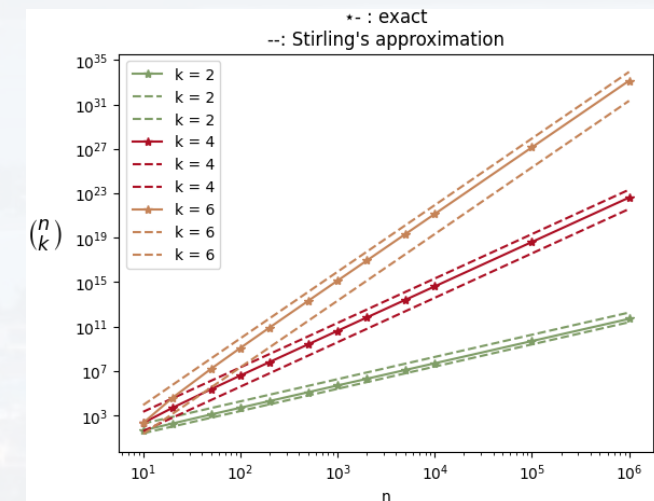
$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{en}{k}\right)^k$$

n choose k

The Binomial Distribution

see `nchoosek.ipynb`

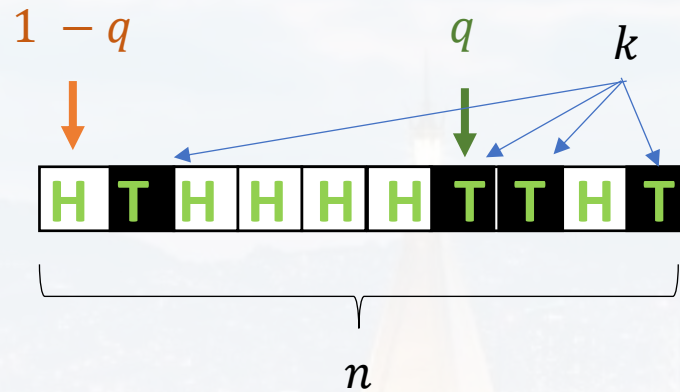
n choose k



n choose k
The Binomial Distribution

bingo = “two”

probability of having a sequence of k tails and $n - k$ heads

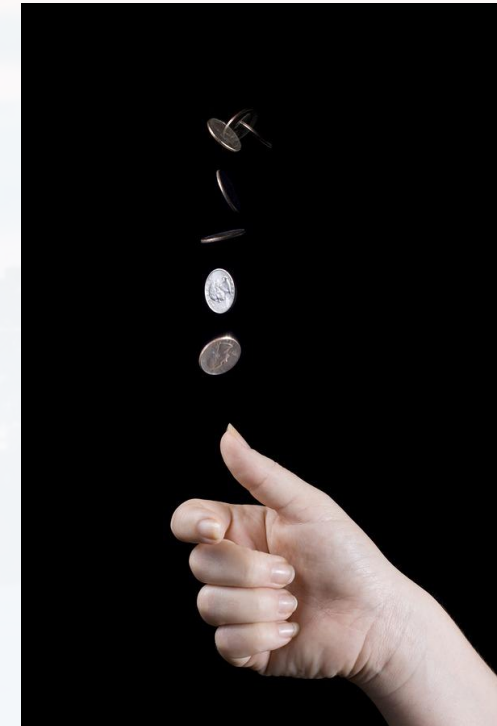


fair coin? $q = 0.5$???

$$p_{tot} = (1 - q)q(1 - q)(1 - q)(1 - q)(1 - q)qq(1 - q)q$$

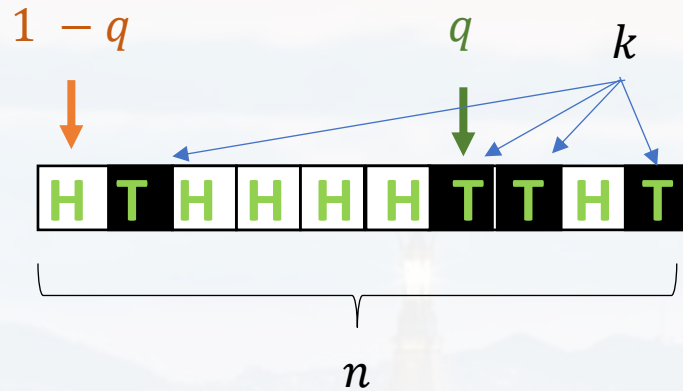
$$p_{tot} = q^k (1 - q)^{n-k}$$

Probability of having **any** sequence of k tails and $n - k$ heads?

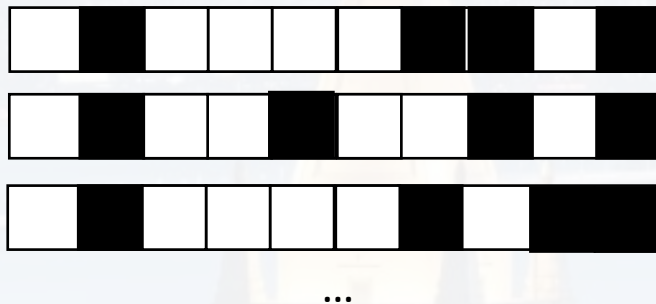


Probability of having **any** sequence of k tails and $n - k$ heads?

n choose k
 The Binomial Distribution



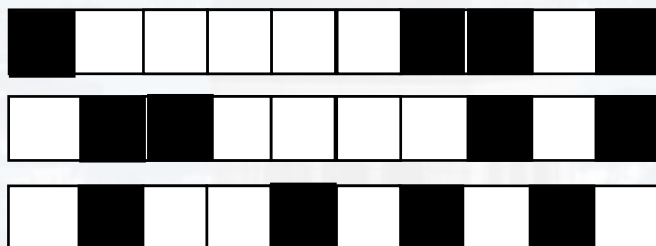
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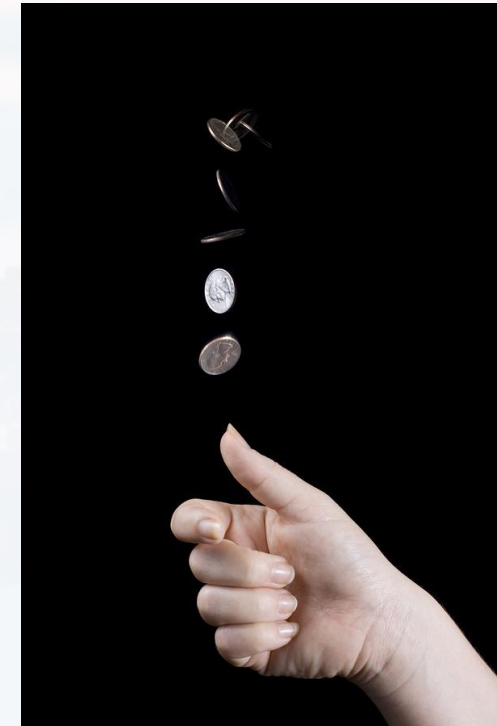
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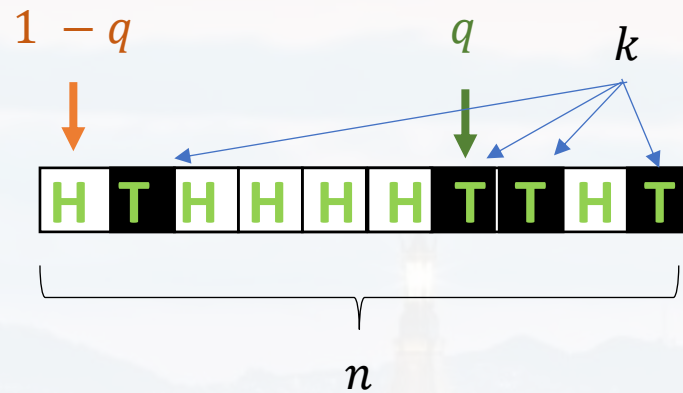
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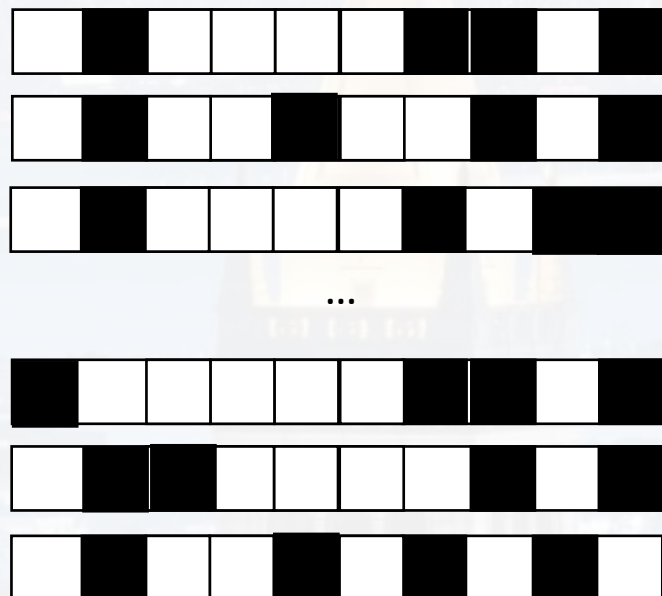
$$p_{tot} = q^k (1 - q)^{n-k}$$



Probability of having **any** sequence of k tails and $n - k$ heads?



$$p_{tot} = q^k (1 - q)^{n-k}$$

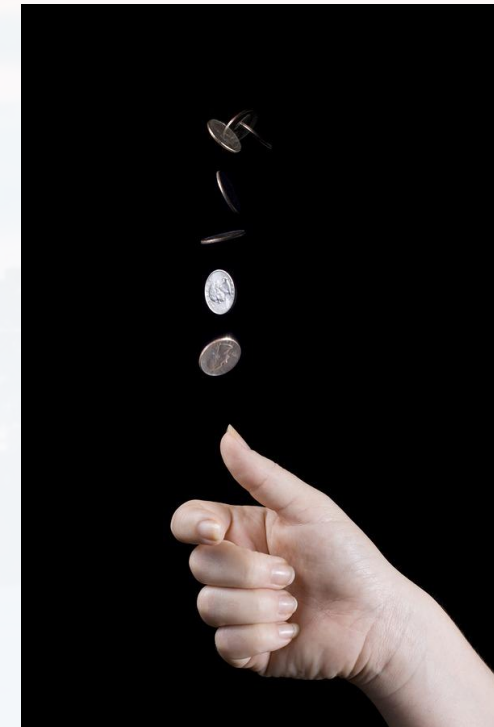


k are *indistinguishable*

$$\Omega = \frac{n!}{k! (n - k)!} = \binom{n}{k}$$

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

n choose k
The Binomial Distribution



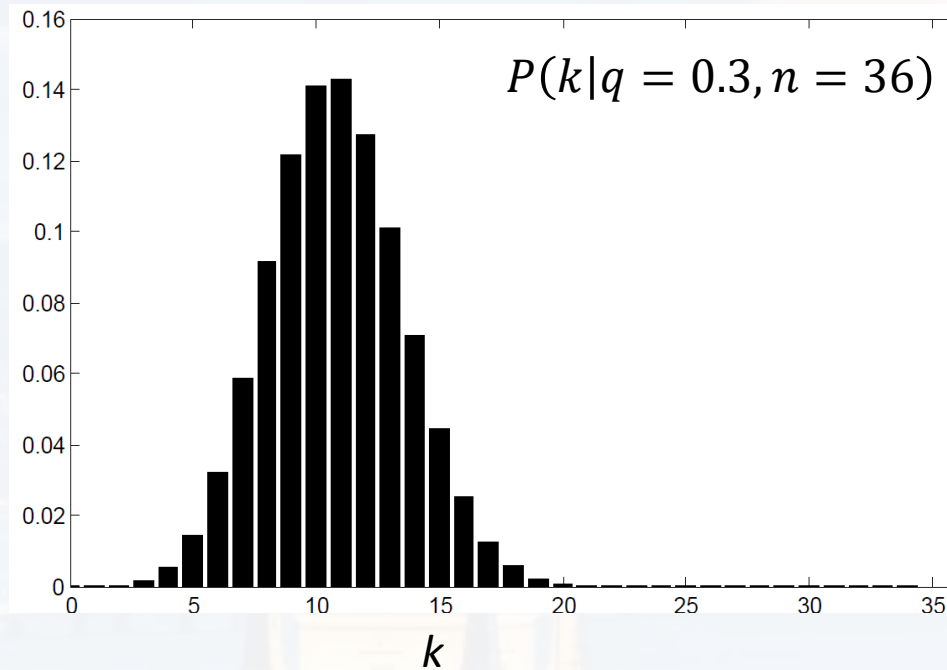
binomial distribution

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

n choose k

The Binomial Distribution



$$\mu = \sum_i x_i p(x_i)$$

$$\mu = \int x p(x) dx$$

$$\mu = \sum_{k=0}^n k \binom{n}{k} q^k (1 - q)^{n-k} = qn$$

$$\text{var}(k) = \sum_{k=0}^n (k - qn)^2 \binom{n}{k} q^k (1 - q)^{n-k} = qn(1 - q)$$

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

n choose k

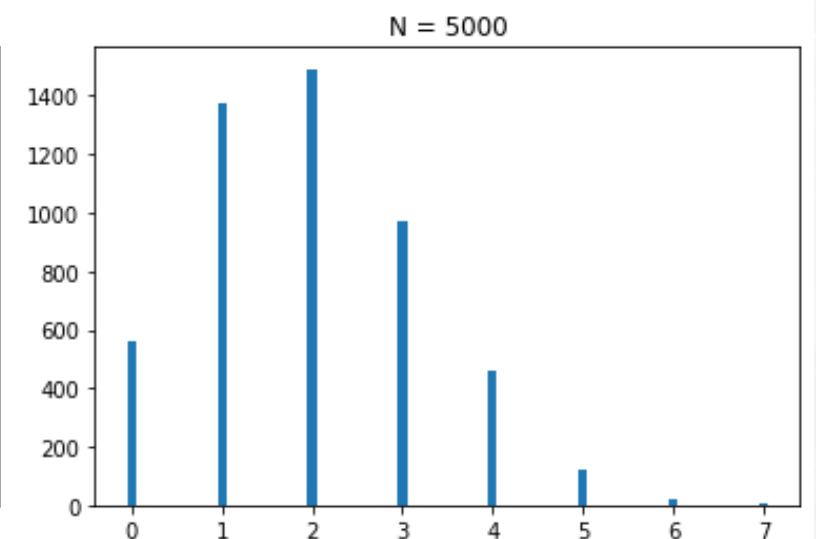
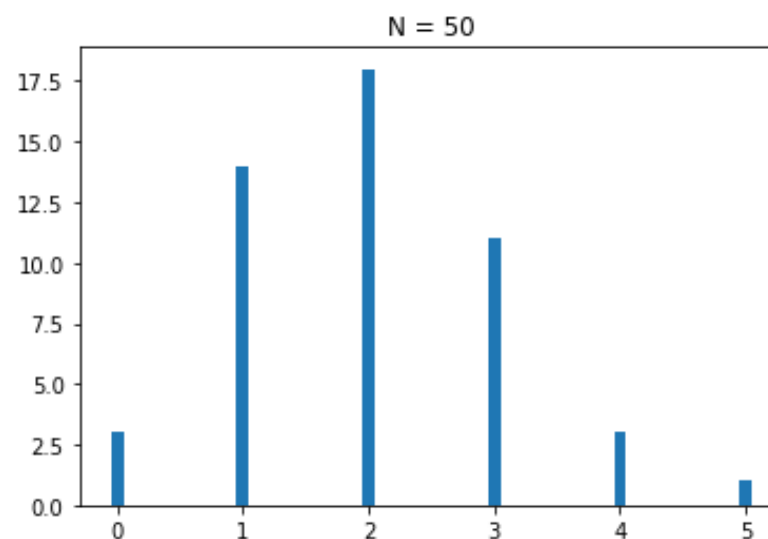
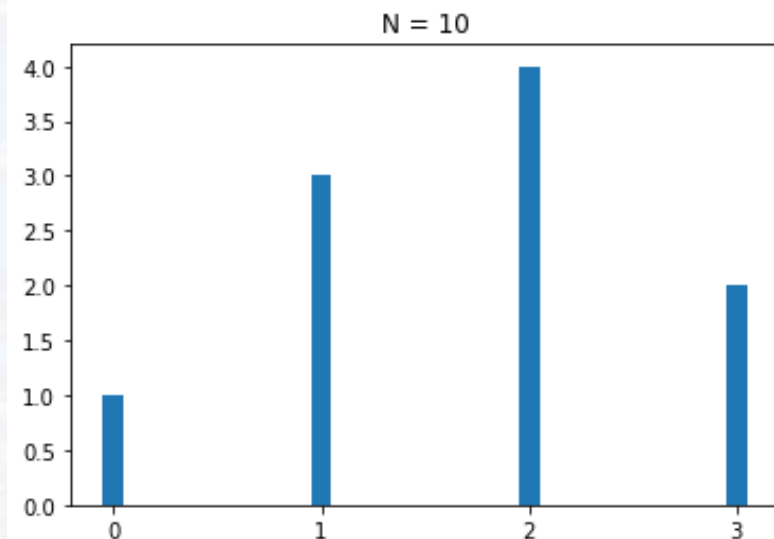
The Binomial Distribution

q = 0.2

n = 10

```
K = np.random.binomial(n, q, N)
```

```
labels, counts = np.unique(K, return_counts = True)
plt.bar(labels, counts, align = 'center', width = 0.1)
plt.gca().set_xticks(labels)
plt.title('N = ' + str(N))
```

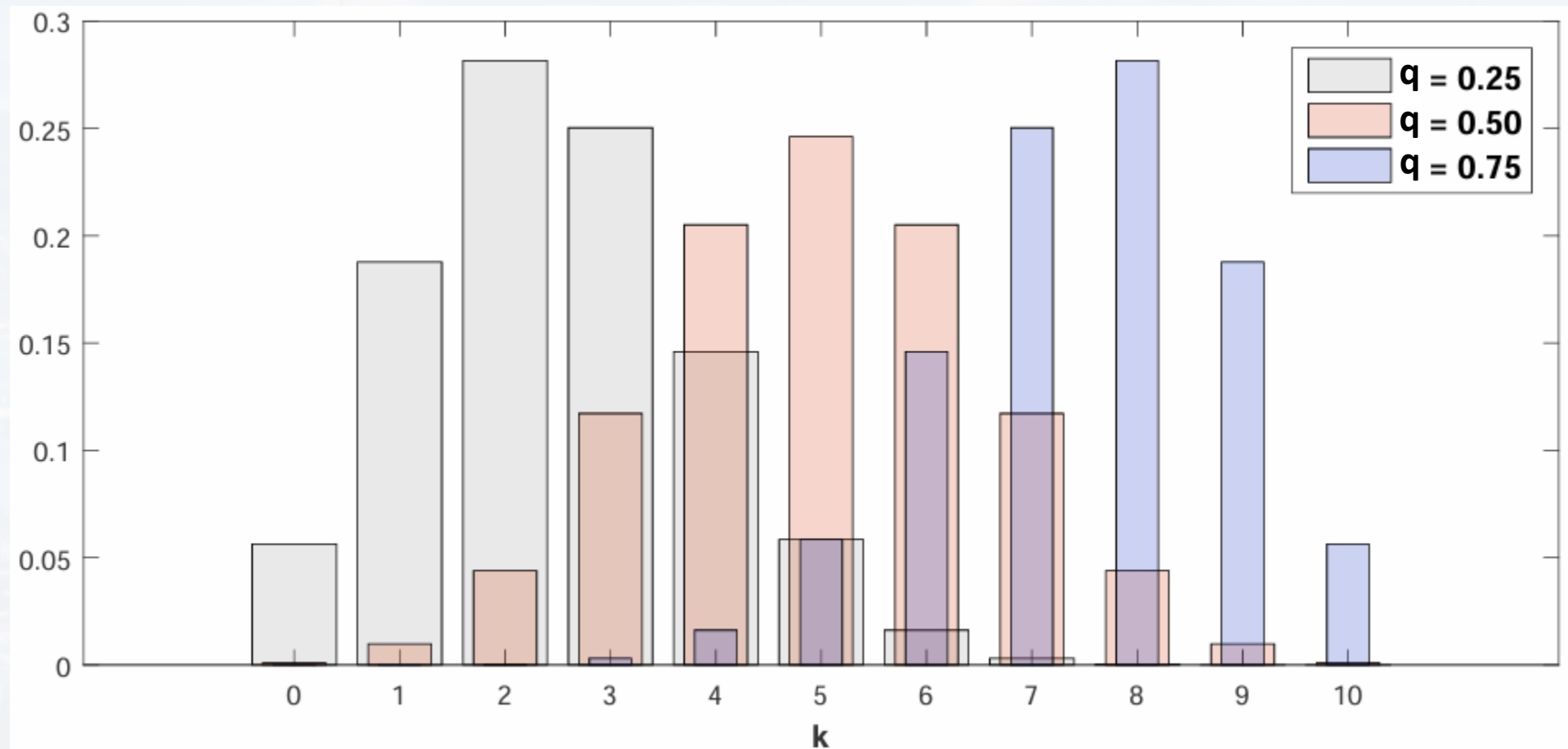


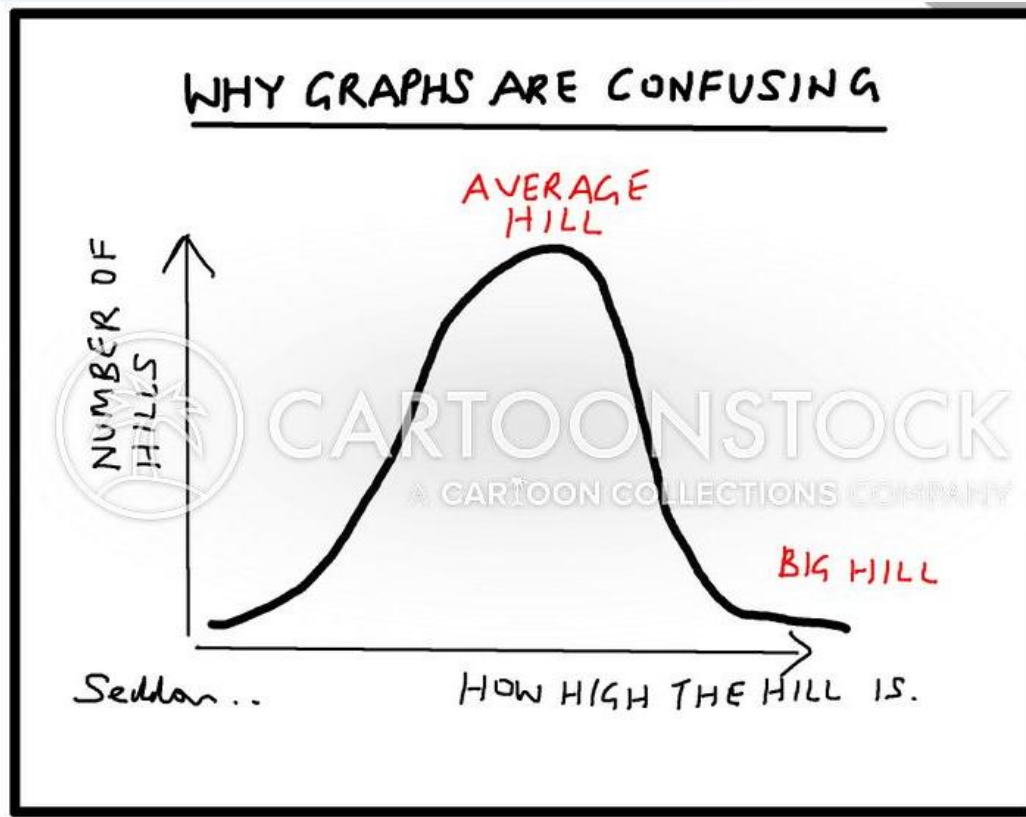
$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

n choose k

The Binomial Distribution





Outline

- Uniform Distribution
- Binomial Distribution
- **Poisson Distribution**
- Normal Distribution
- Central Limit Theorem

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

rare events $\rightarrow q \ll 1$

Taylor expansion for $(1 - q)^{n-k}$ around $q = 0$

$$(1 - q)^{n-k} = 1 - nq + \frac{(nq)^2}{2} - \frac{(nq)^3}{6} + \dots = e^{-nq}$$

$\rightarrow n \rightarrow \infty$

Stirling's approximation for $n!$

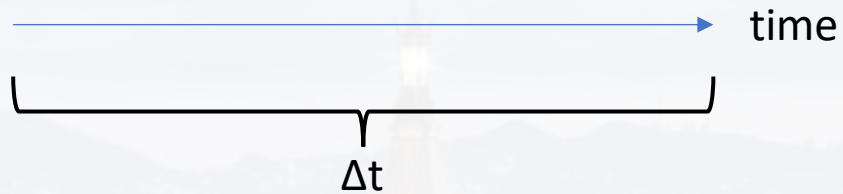
$$\frac{n!}{(n-k)!} \approx \sqrt{\frac{n}{n-k}} \frac{n^n e^{n-k}}{e^n (n-k)^{n-k}} \approx n^k$$

$$\binom{n}{k} q^k (1 - q)^{n-k} \approx \frac{(nq)^k e^{-nq}}{k!}$$

$$\binom{n}{k} q^k (1 - q)^{n-k} \approx \frac{(nq)^k e^{-nq}}{k!}$$

often: $nq := \lambda$

events per time interval: $\lambda = c \Delta t$



rate $c = 4$ tails per Δt



$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$\mu = qn \rightarrow qn = \lambda$$

$$\text{var}(k) = qn(1 - q) \rightarrow qn = \lambda$$



$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$\mu = \lambda$$

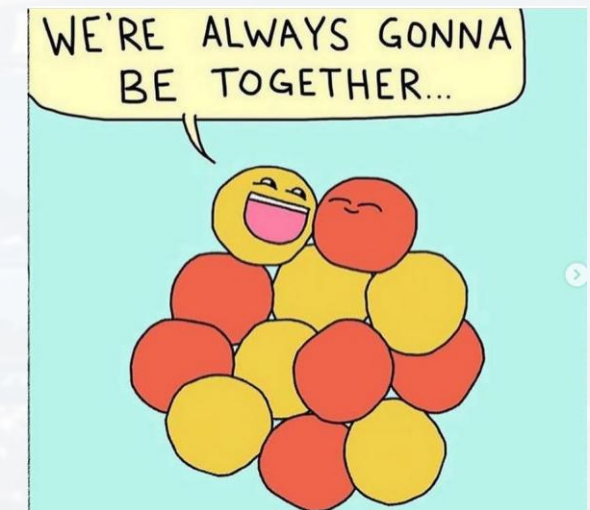
$$\text{var}(k) = \lambda$$

- **rare** events
- events are mutually **independent**
- events have **no duration**

examples:

- radioactive decay
- single photon detection
- lightning
- mutation of a gene
- receiving WhatsApp messages/SMS

rare: not that **a** atom decays,
→ that **this** atom decays within Δt





$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$\mu = \lambda$$

$$\text{var}(k) = \lambda$$

```
c      = 5  
delt   = 10  
lam    = c * delt
```

```
K      = np.random.poisson(lam, N)
```

```
labels, counts = np.unique(K, return_counts = True)  
plt.bar(labels, counts, align = 'center', width = 0.1)  
plt.gca().set_xticks(labels)  
plt.title('N = ' + str(N))
```



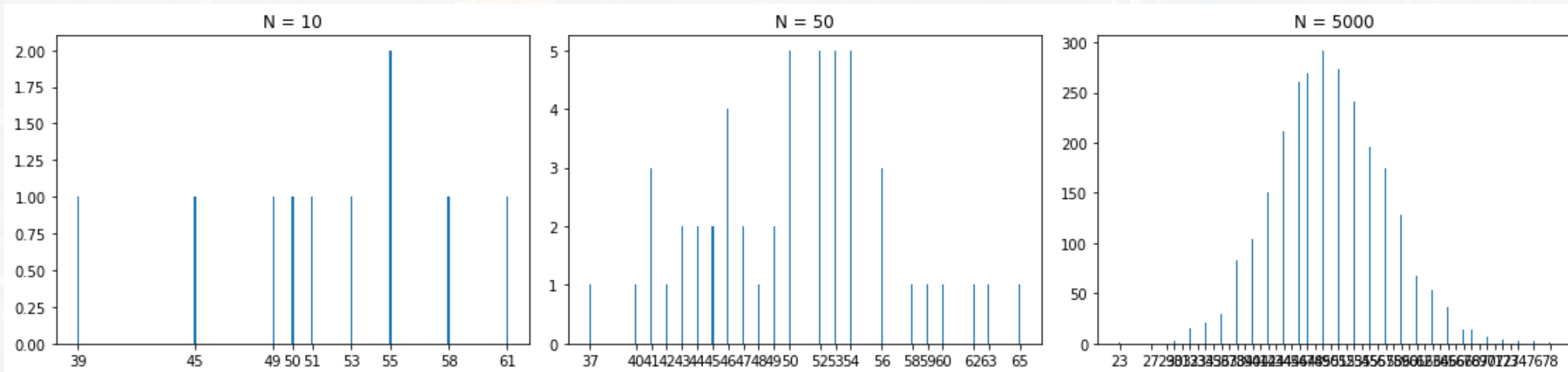
$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

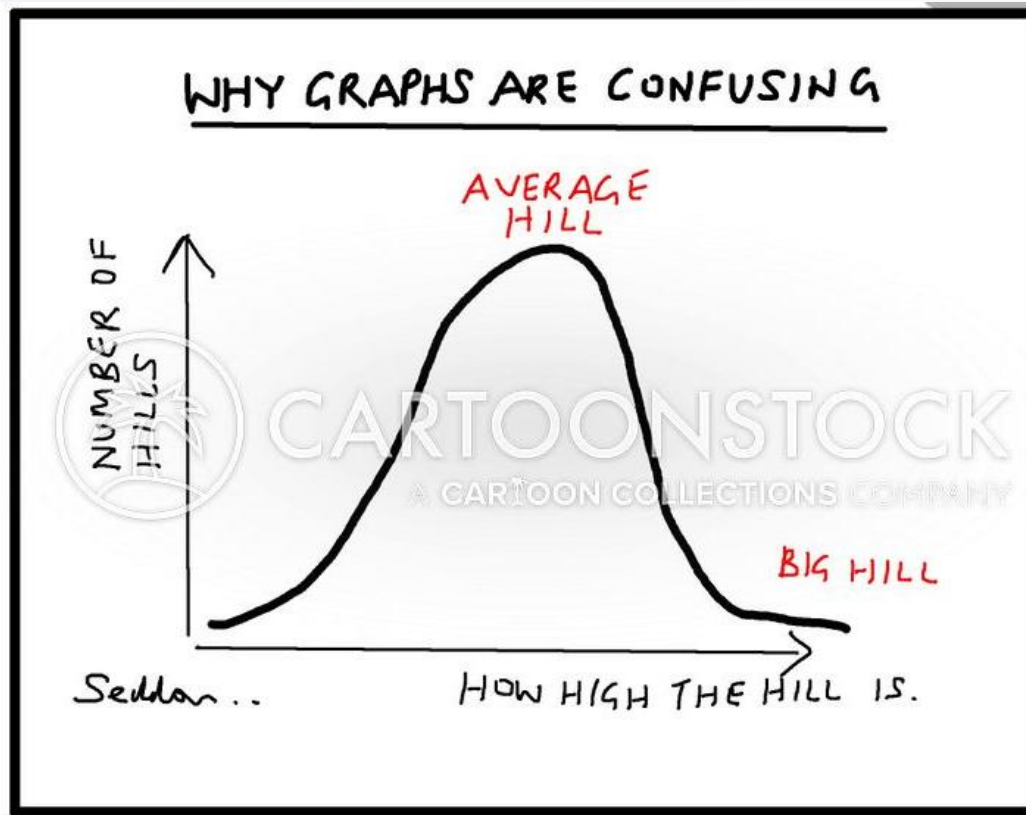
Poisson distribution

$$\mu = \lambda$$

$$\text{var}(k) = \lambda$$

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```





Outline

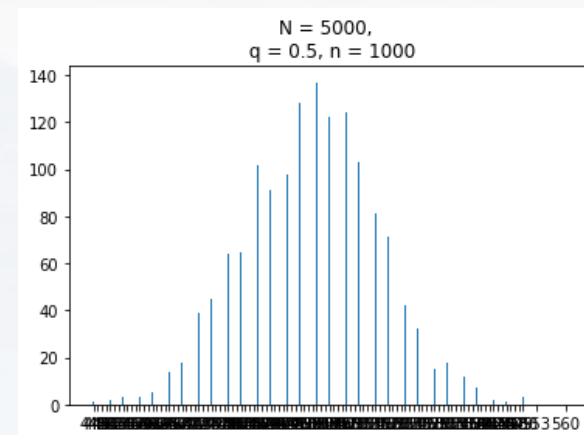
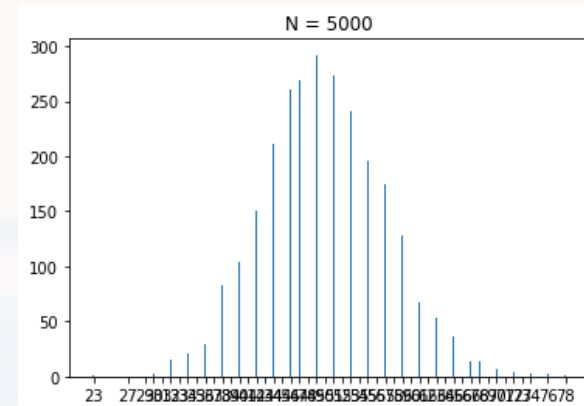
- Uniform Distribution
- Binomial Distribution
- Poisson Distribution
- **Normal Distribution**
- Central Limit Theorem

$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution



Stirling's approximation for even larger n

$$P(k|n, p) \approx \frac{1}{\sqrt{2\pi nq(1-q)}} \exp\left[-\frac{(k - nq)^2}{2nq(1-q)}\right]$$



Stirling's approximation for even larger n

$$P(k|n, p) \approx \frac{1}{\sqrt{2\pi nq(1-q)}} \exp\left[-\frac{(k-nq)^2}{2nq(1-q)}\right]$$

using $\sigma^2 = \text{var}(k) = qn(1-q)$

$$\mu = qn$$

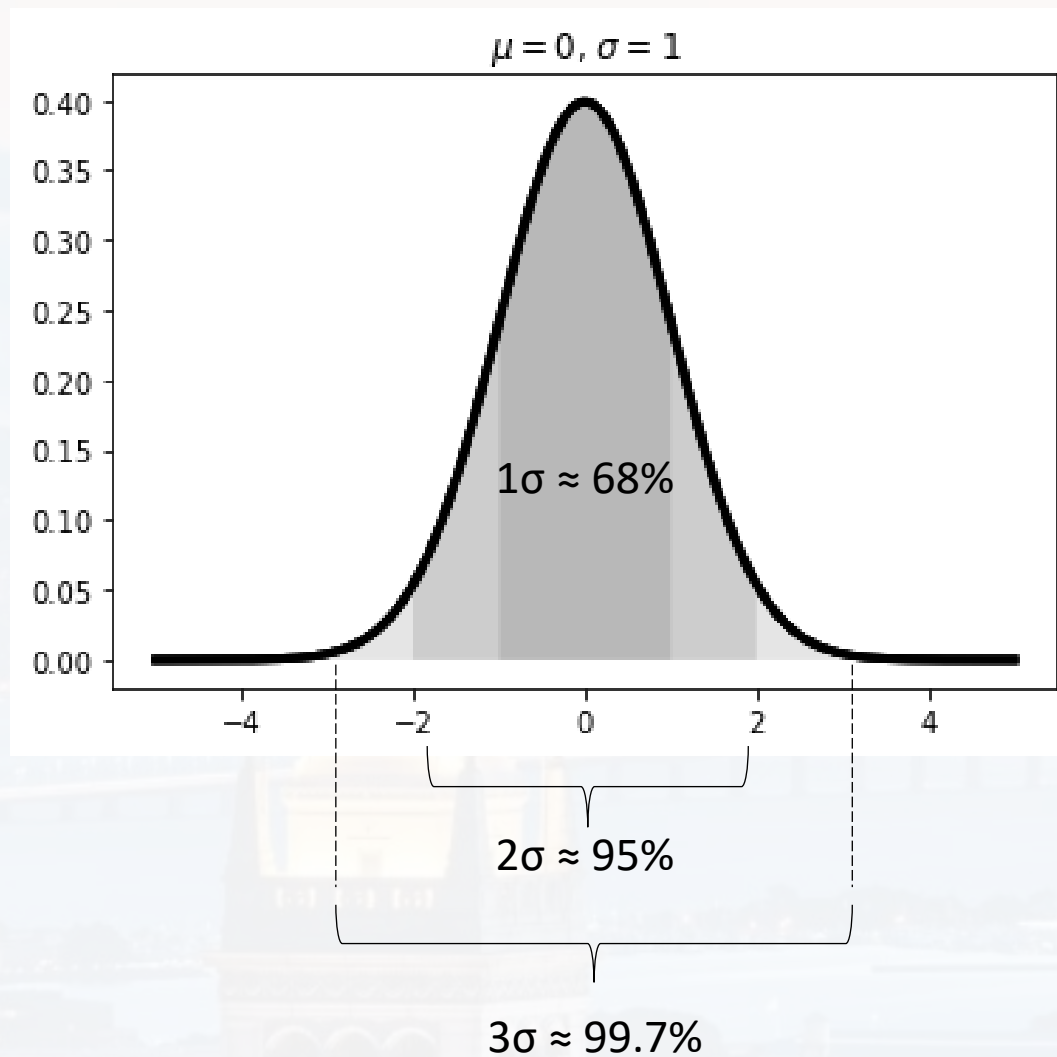
and $k := x$

$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi \sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2 \sigma^2}\right]$$

Normal/Gauss distribution

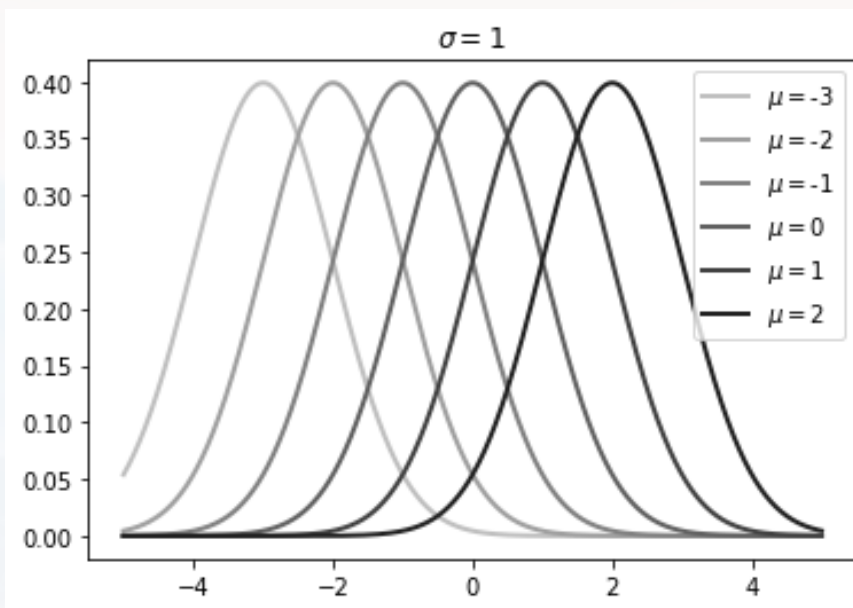


Note, that the **Poisson** and the **Binomial distribution** are *discrete*,
whereas the **Normal distribution** is *continuous*!



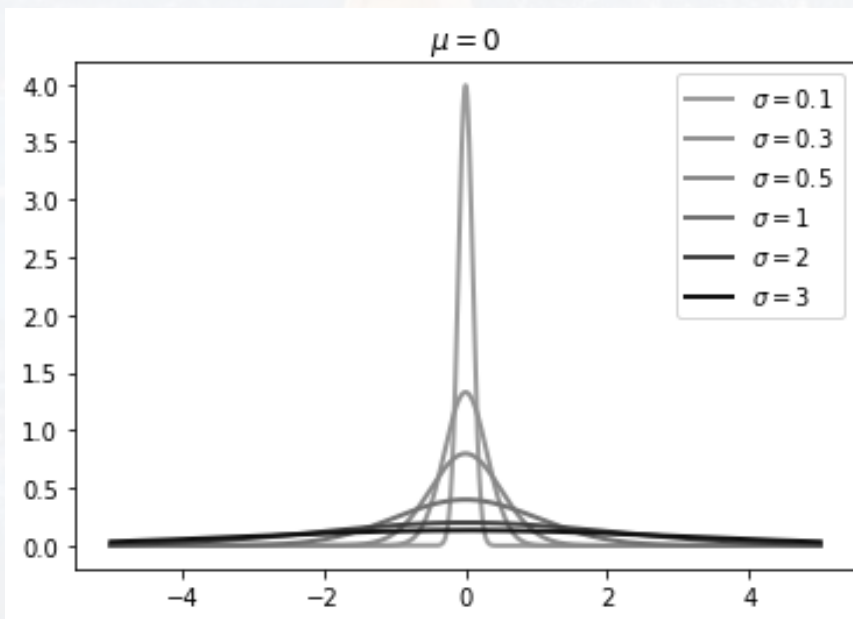
$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

Normal/Gauss distribution



$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

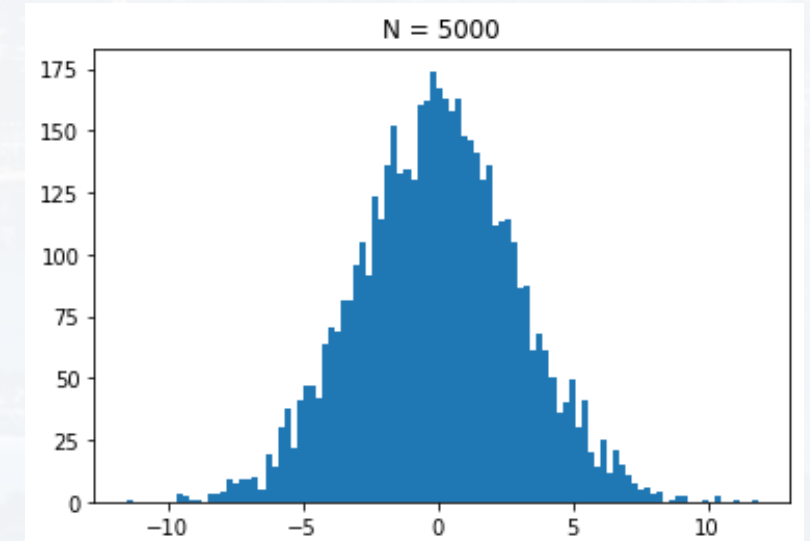
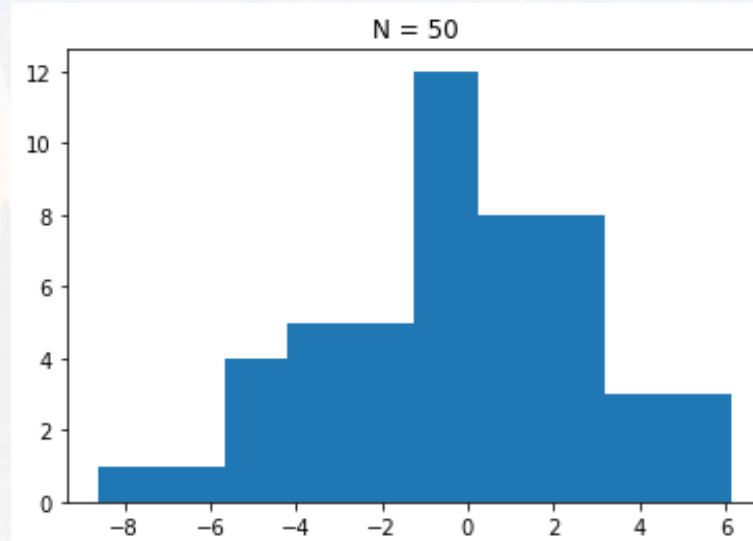
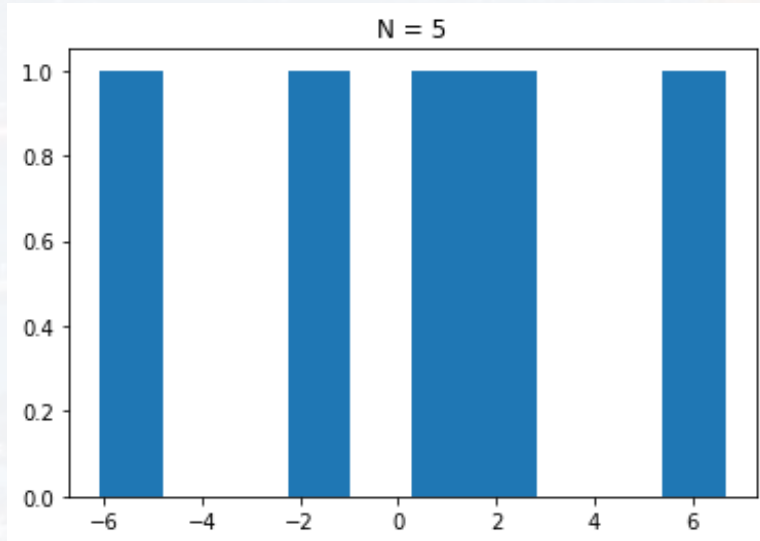
Normal/Gauss distribution

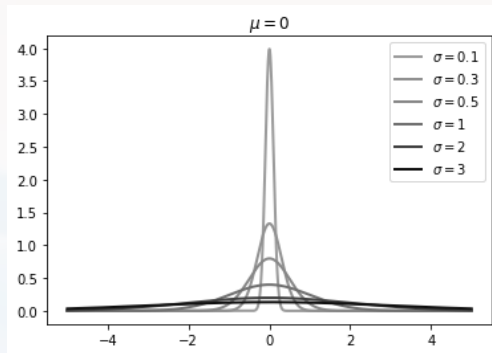


```
mu = 0  
s = 1  
P = np.random.normal(mu, s, N)  
plt.hist(P)  
plt.title('N = ' + str(N))
```

$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

Normal/Gauss distribution





$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

Normal/Gauss distribution

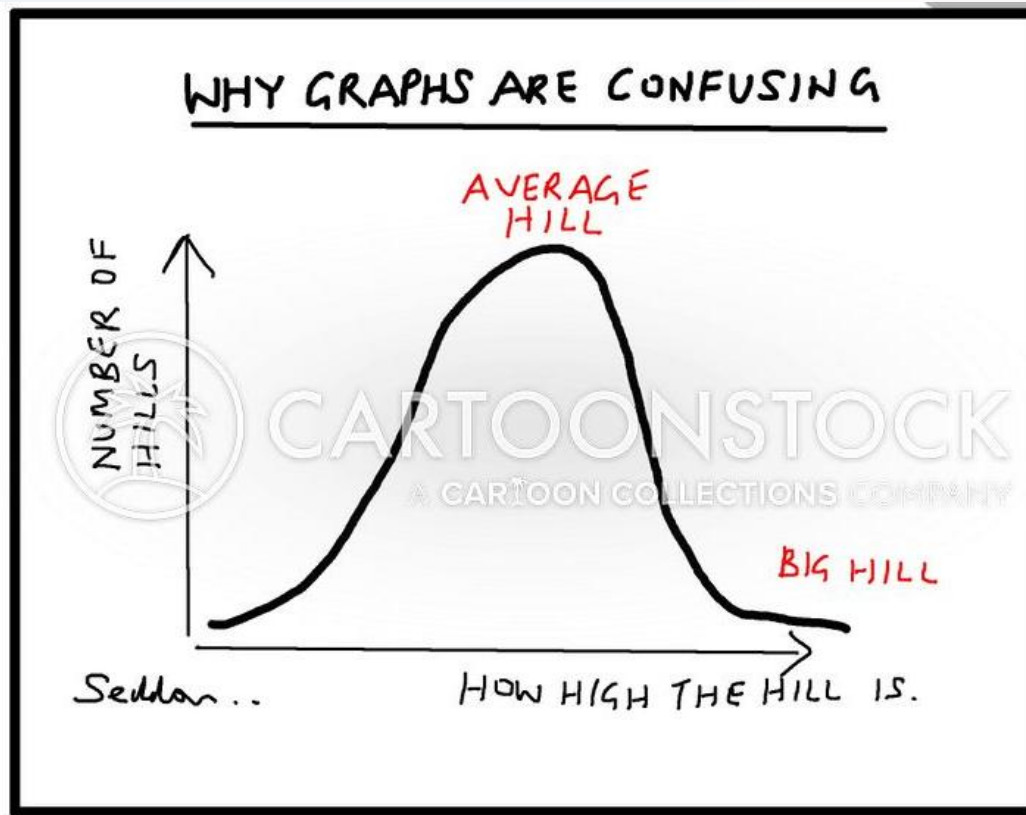
examples:

- diffusion processes
- approx. stat. error of data points
- approx. distribution of body height/shoe sizes/ weight, IQ
- approx. blood pressure, blood values
- approx. retirement age
-

applications:

- significance tests
- t-test
- ANOVA/MANOVA
- χ^2 - test
- χ^2 itself and students-t distribution
- ...

Why do so many quantities follow a normal distribution?



Outline

- Uniform Distribution
- Binomial Distribution
- Poisson Distribution
- Normal Distribution
- **Central Limit Theorem**



binomial distribution

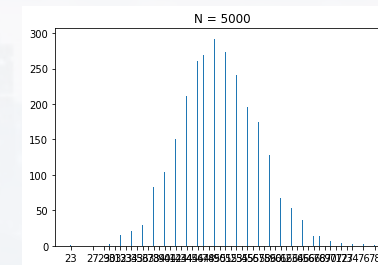
$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

$q \rightarrow 0$

Poisson distribution

$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

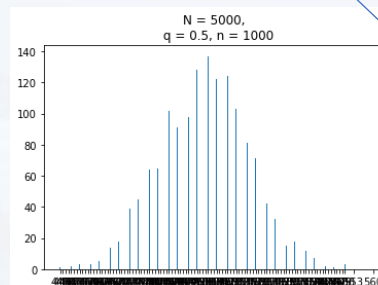
$n \rightarrow \infty$



$n \rightarrow \infty$

The fact that that many datasets can be well approximated by a normal distribution for $n \rightarrow \infty$ is called

Central Limit Theorem



Normal/Gauss distribution

$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma^2} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$



Why do so many quantities follow a normal distribution?

At the end... all probability distributions are **Maximum Entropy** Distributions, subject to a **set of constraints**

Distribution name	Probability density / mass function	Maximum Entropy constraint	Support
Uniform (discrete)	$f(k) = \frac{1}{b - a + 1}$	None	$\{a, a + 1, \dots, b - 1, b\}$
Uniform (continuous)	$f(x) = \frac{1}{b - a}$	None	$[a, b]$
Bernoulli	$f(k) = p^k (1 - p)^{1-k}$	$\mathbb{E}[K] = p$	$\{0, 1\}$
Geometric	$f(k) = (1 - p)^{k-1} p$	$\mathbb{E}[K] = \frac{1}{p}$	$\mathbb{N} \setminus \{0\} = \{1, 2, 3, \dots\}$
Exponential	$f(x) = \lambda \exp(-\lambda x)$	$\mathbb{E}[X] = \frac{1}{\lambda}$	$[0, \infty)$
Laplace	$f(x) = \frac{1}{2b} \exp\left(-\frac{ x - \mu }{b}\right)$	$\mathbb{E}[X - \mu] = b$	$(-\infty, \infty)$
Asymmetric Laplace	$f(x) = \frac{\lambda \exp\left(-(x - m) \lambda s \kappa^s\right)}{\left(\kappa + \frac{1}{\kappa}\right)}$ where $s \equiv \text{sgn}(x - m)$	$\mathbb{E}[(X - m) s \kappa^s] = \frac{1}{\lambda}$	$(-\infty, \infty)$
Pareto	$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$	$\mathbb{E}[\ln X] = \frac{1}{\alpha} + \ln(x_m)$	$[x_m, \infty)$
Normal	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$	$\mathbb{E}[X] = \mu,$ $\mathbb{E}[X^2] = \sigma^2 + \mu^2$	$(-\infty, \infty)$



Why do so many quantities follow a normal distribution?

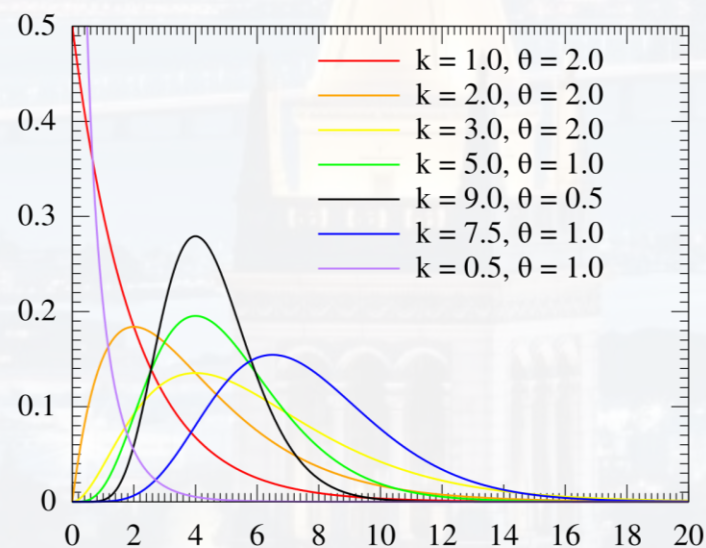
At the end... all probability distributions are **Maximum Entropy** Distributions, subject to a **set of constraints**

examples:

- approx. stat. error of data points
- approx. distribution of body height/shoe sizes/ weight, IQ
- approx. blood pressure, blood values
- approx. retirement age

....

Gamma	$f(x) = \frac{x^{k-1} \exp\left(-\frac{x}{\theta}\right)}{\theta^k \Gamma(k)}$	$\begin{aligned} \mathbb{E}[X] &= k\theta, \\ \mathbb{E}[\ln X] &= \psi(k) + \ln \theta \end{aligned}$	$[0, \infty)$
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note:

often, the exact model is actually the **gamma distribution**, but a normal distribution is **simpler** and usually a **good approximation**



Thank you very much for your attention!

